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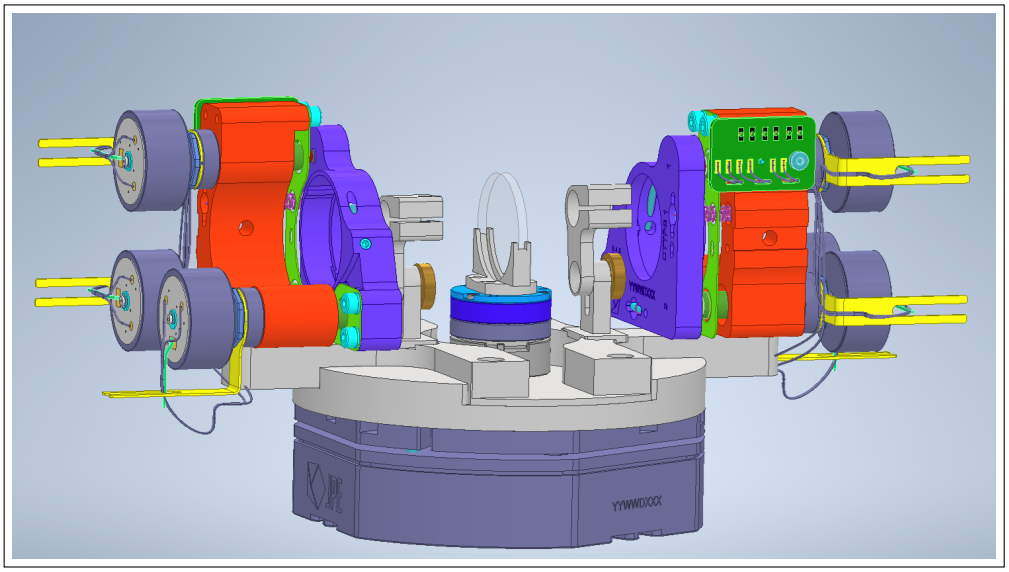
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Optimizing the alignment of a cryogenic optical filter bench

Bachelor Thesis



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Abstract

This thesis presents the development and optimization of a Cryogenic Optical Filter Bench (COFB), a critical component in high precision astroparticle physics experiments. The COFB is designed to suppress blackbody photon signals utilizing a narrowband optical filter, while preserving high detection efficiency. By utilizing two tip/tilt piston stages and implementing an iterative method to scan a six-parameter space, the system achieved consistent and reproducible transmission above 50%. The primary challenges related to lens collimation and beam alignment are outlined, and strategies for overcoming alignment issues caused by internal reflections or diffraction effects are discussed. Finally, a brief analysis addresses the challenges of cryogenic deployment, highlighting which issues the current COFB system can handle and identifying areas requiring further hardware or software development.

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List of Acronyms

ALPs Axion-Like-Particles. 4

ALPS II Any Light Particle Search collaboration. 6–9, 11–13, 39, 40

CLA Cryo Linear Actuators. 20, 25

COFB Cryogenic Optical Filter Bench. 4, 13–21, 23–26, 30, 34, 36, 38–42

DCR Dark-Count Rate. 9, 11, 13, 39, 40

DESY Deutsches Elektronen-Synchrotron. 6

DQL Dark Quantum Lab. 15, 22, 23

ePS emitting piston stage. 27–29, 31–33, 37, 40–42

FWHM Full Width at Half Maximum. 10, 21, 37

rPS receiving piston stage. 27–42

SDE System Detection Efficiency. 9, 14, 42

SDU University of Southern Denmark. 15, 25

SNSPD Nanowire Single-Photon Detector. 9, 11, 13, 24

SQUID Superconducting Quantum Interference Device. 9, 10

TES Transitions-Edge Sensor. 9–11, 13, 16

Chapter 1

Introduction to astroparticle physics, the missing mass problem and axions

Since the dawn of mankind, humans have gazed into the night sky, contemplating its contents. Then in the seventeenth century, the first telescopes allowed researchers to explore the solar system, leading to the paradigm-shifting realization that the Sun was the centre of the solar system. As telescopes improved discoveries quickly followed. Suddenly the sun was not only the centre of the solar system, but just one of billions and billions of stars in the Milky Way, which in turn was just a singular speck of dust in a vastness so incomprehensible it redefined our very understanding of existence itself.

This discovery gave rise to a multitude of astronomical research branches, from cosmology, the study of the origin of the universe, and astrophysics, applying well-tested physical theory to astronomical objects, to astrobiology, the study of life in the universe, and archaeoastronomy, the study of the impact of the universe on early human civilizations. As scientists on Earth pushed the boundaries of our understanding, the field of particle physics, the study of the fundamental building blocks of matter and energy, branched out and created the interdisciplinary field of astroparticle physics. This research field bridges the fundamental building blocks, also known as particles, with astronomical observations, enabling scientists to study cosmic rays, active galactic nuclei, supernovae and other celestial phenomena through the interactions and behaviour of particles.

Then in 1933, the Swiss astronomer Fritz Zwicky was studying the Coma galaxy cluster and discovered a discrepancy in the galaxy's rotational velocity and their observed mass [1, 2]. Zwicky postulated that this additional velocity could be explained by increasing the masses of the galaxies, a mass he termed *dunkle Materie*. This additional mass has since become known as dark

matter. Overwhelming evidence for dark matter has since been discovered from various astrophysical measurements, such as galaxy rotation curves, gravitational lensing or the cosmic microwave background [3]. Today, physicists estimate that 4.84% of the universe is made from *baryonic* matter, described by particles from the standard model, 25.8% is the elusive dark matter, and the remaining 69.2% is an unknown substance known to as dark energy [4]. Stars, planets and gas are believed to make up about 6% of the baryonic matter in the universe.

The nature of dark matter is still perplexing scientists, and several new particles have been proposed to solve this mystery. The hypothetical particles should be *weakly interacting*. In other words, these *dark matter candidates* should not affect each others, or other particles, behaviour or properties. An example of such a hypothetical particle is the Weakly-Interacting-Massive-Particles (WIMPs), which don't interact with light, but still have mass to interact through gravity [5].

Another dark-matter candidate is the axion. Axions, or other Axion-Like-Particles (ALPs), have smaller mass than WIMPs, but interact with light and should therefore be detectable by scientists on Earth, or using telescopes in orbit. Additionally, since these axions interact with light, they could potentially be produced in a laboratory on Earth.

To detect axions on Earth, new technology needs development. Apart from super sensitive detection devices to detect light particles, known as photons, from axion interactions, filters to distinguish light from these interactions and light from environmental sources are also in development. The contents of this thesis describe the preliminary work required to implement a filter technology to an axion-detecting system efficiently. For the filter to work as intended, it must be positioned at a very low temperature, also known as a cryogenic environment. To realise this, the filter must be placed on a platform, which can support adjustments, even at cryogenic temperatures. The cryogenic temperature environment is realised by inserting the filter-bench into a dilution refrigerator, that uses helium to reach temperatures below $-272°$. The combined term for the platform and filter technology is known as a Cryogenic Optical Filter Bench (COFB), and the following chapters detail the process of designing and implementing effective control of such a system, and the initial results from operating the system at room temperature.

Chapter 2

Theoretical Foundations for Axion Searches with ALPS II

2.1 Axions and Axion-like-particles

Axions are hypothetical particles originally proposed by R. D. Peccei and H. R. Quinn in 1977 [6, 7] to solve the charge-conjugation and parity (CP) violation problem in Quantum Chromodynamics (QCD), known as the "Strong CP" problem [8, 9]. Axions has since also been considered as a dark-matter candidate [10].

While axions has not yet been observed, multiple production channels are believed to exist [11, 12, 13]: *relic axions* produced during the early universe, *solar axions* produced in stars and axions produced in high-energy environments such as supernovae. The primary mechanism of solar axion production is the Primakoff effect [14] and is described by the Lagrangian coupling term between the electromagnetic field interaction of two propagating photons and the axion field as

$$\mathcal{L} = g_{a\gamma\gamma} a \mathbf{E} \cdot \mathbf{B} = -\frac{g_{a\gamma}}{4} a F_{\mu\nu} \tilde{F}^{\mu\nu} \tag{2.1.1}$$

where $g_{a\gamma\gamma}$ is the axion-photon coupling constant, a is the axion field, $\mathbf{E}$ the electric field of the photon, $\mathbf{B}$ a external magnetic field and $F_{\mu\nu}$ the electromagnetic field tensor (and its dual $\tilde{F}^{\mu\nu}$). As photons are produced in stars, they interact with charged particles in the plasma and convert into axions which could potentially be observed on Earth. To observe axions the particles would have to re-convert into photons. This is believed to be achievable by utilizing the inverse Primakoff effect, but instead of interacting with charged particles, a static magnetic field would be applied providing the necessary conditions for the axions to convert into photons.

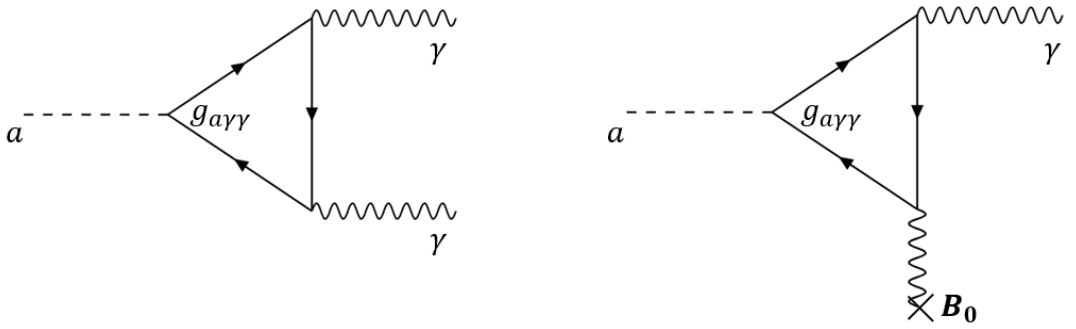


Figure 2.1.1: Feynman diagrams for the inverse Primakoff effect. Left: one-loop diagram. Right: with a virtual photon from a static magnetic field.

Furthermore, by shooting photons into cavities with strong magnetic fields, it is believed that axions could be created in the laboratory, which could then be detected [15]. In this thesis, the focus will be on the axion search covered by the Any Light Particle Search collaboration (ALPS II).

2.2 ALPS II

The ALPS II collaboration is a research group located at the Deutsches Elektronen-Synchrotron (DESY) in Hamburg. The ALPS II experiment is a so-called *light-shining through walls* experiment [16, 17], in which a high power laser shoots a beam of photons into the production cavity at an opaque wall. In the cavity, the photons would be subject to a high magnetic field, produced by superconducting magnets, and be converted through the Primakoff effect into axions [18]. These axions then pass through the wall, as they are very weakly interacting with matter, and enter the regeneration cavity. In the regeneration cavity, the axions would then convert back into photons which are collected and sent through an optical system into a single-photon detector [19].

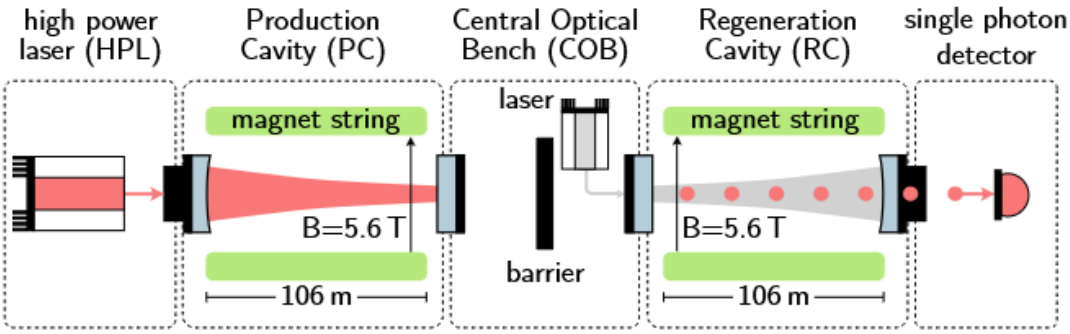


Figure 2.2.1: Sketch of the ALPS II experiment from [19]: "The Any Light Particle Search Experiment at DESY".

The governing factor in the amount of detected photons is the coupling factor and the mass of the axion. The mass of the axion is bound by the coherence to $m_a \leq 10^{-4}$eV, while the ALPS II experiment aims to probe a coupling coefficient of $g_{a\gamma\gamma} = 2 \cdot 10^{-11}\text{GeV}^{-1}$ [18]. The probability of photon-axion conversion at these parameters in a cavity is [11]

$$\mathcal{P}(a \to \gamma) \sim \left(\frac{g_{a\gamma\gamma}BL}{2}\right)^2 \tag{2.2.1}$$

Since the axions would then have to convert into photons in the regeneration cavity, the total probability of the photon-axion-photon conversion process is the square of the probability in (2.2.1). The probability of the complete photon-axion-photon conversion is minuscule ($\sim 1 \cdot 10^{-21}$) and without modifying the process it is unfeasible to prove the existence of axions. To achieve a higher conversion factor positioning mirrors into the production- and regeneration cavities 2.2.1, effectively trapping the beam and increasing the photon density, introduces a power build-up factor such that the total photon count at the detector is increased [19]

$$N_\gamma = \left(\frac{g_{a\gamma\gamma}BL}{2}\right)^4 \cdot \eta\beta_{\text{RC}} \cdot P_{\text{PC}} \cdot \tau \tag{2.2.2}$$

where β_{RC} ($= 40,000$) and P_{PC} ($= 150$kW) is the power buildup factors (for a photon beam with $\lambda = 1064$nm), τ is the observation time and η the detection efficiency. The expected photon signal int ALPS II experiment is 1 photon per day and the main issues with a low photon rate are the sensitivity of the detector to dark counts and the stability of the detector over long measurement periods. In the ALPS II experiment the dark count rate must not exceed 7μHz (one dark count every ~ 1.65 days) to achieve significant evidence of axions while still maintaining high detector efficiency [19].

Chapter 3

Experimental Framework for Cryogenic Single Photon Detectors and Optical Filter Benches with ALPS II

This chapter presents the experimental framework of cryogenic single-photon detectors and optical filter benches, which have been planned to operate in parallel to the current heterodyne detection scheme at the ALPS II experiment. These systems are critical for achieving low dark-count rate while maintaining high detection efficiency required to detect the extremely rare photon signals potentially associated with the detection of axion.

3.1 Cryogenic single-photon detectors

To reduce the effect of background signals, discussed in the following sections, the detection system is built around cryogenic single-photon detectors (SPD), rather than room-temperature SPDs. Cryogenic single-photon detectors utilize superconducting materials biased (current or voltage) close to their superconducting transition temperature where the resistance of the material changes steeply from zero to normal state R as a function of temperature [20]. This change is so steep that even the absorption of a single photon heats the material sufficiently to cause a remarkable change in its resistance. A superconducting material has zero electrical resistance and electrons flow freely through the material creating a substantial current without resistive power losses. For this effect to occur, the material must be below a certain temperature, external magnetic field and current density, all dependent on each other. Then as a photon is absorbed by the system, it causes a slight temperature change, which in turn causes a significant increase

in electrical resistivity [21]. As part of the ALPS II experiment, two main detection technologies are currently being investigated: the Transitions-Edge Sensor (TES) and the Nanowire Single-Photon Detector (SNSPD)*.

The viability of cryogenic single photon detection is based on several characteristics. In this thesis the indistinguishability of incident photons arriving at the detector from true signals, and detection events without any real photons incident on the detector, known as dark counts, are detailed, and a method to reduce dark counts using optical filters is discussed. The principal figures of merit for single-photon detectors, relevant to this thesis, are:

1. System Detection Efficiency (SDE): The probability of detecting an incident photon taken into account all optical losses and quantum efficiency of the sensor.
2. Dark-Count Rate (DCR): The rate (Hz) of dark counts that are indistinguishable from the light signals given the energy resolution of the sensor.

Dark counts (DC) are subdivided into intrinsic- and extrinsic dark counts. Intrinsic dark counts are triggers in the cryogenic single photon detector without a fibre connection, originating from cosmic rays, radioactive decay and electric noise. The extrinsic dark counts originate from stray and blackbody photons that couple into the optical fibre linking the room-temperature laboratory environment to the single-photon detector inside the dilution refrigerator.

The Transition-Edge Sensor

A TES is a thin film of superconducting material kept below its superconducting transmission temperature voltage-biased close to the superconducting-to-normal transition state [22]. In practice, the TES current is coupled inductively via a magnetic flux to a Superconducting Quantum Interference Device (SQUID), converting the changes in current into a measurable voltage across the SQUID loop. After an incident photon triggers the detector, negative electrothermal feedback from Joule heating returns the system to the superconducting state. The TES are energy-resolving sensors with high quantum efficiency, low dark count rate and microsecond-range dead time [23, 24, 25].

Superconducting Quantum Interference Device

A SQUID is a superconducting loop interrupted by two insulating Josephson junctions where Cooper pairs can tunnel through. This creates an interference effect manifesting as a sinusoidally varying voltage across the SQUID loop as a function of magnetic flux. This makes SQUIDs the most sensitive magnetometers that are well suited for monitoring the TES current [26].

*The SNSPD are not being tested for axion detection.

Transition-edge sensor energy resolution

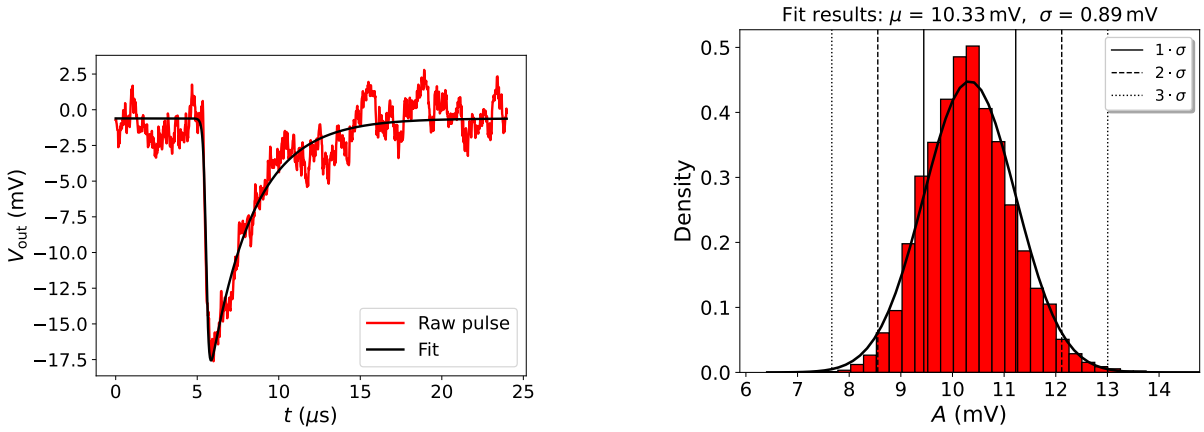


Figure 3.1.1: Left; example voltage pulse of a signal photon incident on a TES, with fit. Right; Example histogram of the pulse amplitudes with Gaussian fit. The energy-resolving detector is characterised such that the mean μ corresponds to the investigated wavelength.

As concluded above, the change in TES current induced by the absorption of a photon is detected by monitoring the SQUID voltage. This results in a pulse, as illustrated in Figure 3.1.1. Traditionally, the energy of the absorbed photon is considered proportional to the integral of the recorded pulse over a wide range of photon wavelengths, derived from a fit to the pulse. However, alternative pulse characteristics are being investigated for their potential equivalence in representing photon energy [27]. By fitting the combined histogram of these fit-derived energy estimators, a probability density function (PDF) is obtained, from which the energy resolution of the TES can be determined. As seen in Figure 3.1.1, a Gaussian fit is employed to analyse the data.

$$f(E_{\text{obs}}, E_{\text{mean}}) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(E_{\text{obs}} - E_{\text{mean}})^2}{2\sigma^2}\right] \tag{3.1.1}$$

where E_{obs} is the energy of an observed photon, E_{mean} is the mean (peak) energy of the photon-pulse data and σ is the standard deviation. The energy resolution can be quantified as the standard deviation σ of the Gaussian fit of a detector, divided by the mean μ of the Gaussian fit,

$$\text{Energy Resolution} = \frac{\sigma}{E_{\text{mean}}}. \tag{3.1.2}$$

The energy resolution is traditionally accompanied by the Full Width at Half Maximum (FWHM) (FWHM $\approx 2.355 \cdot \sigma$), to indicate the width of the energy peak. A careful analysis of the pulses in the frequency domain has resulted in the highest energy resolution of 5.3% (FWHM = 0.15 eV) [28]. A smaller TES energy resolution allows for better distinction between true 1064 nm

signal photons, and background blackbody photons. Thus, by lowering the energy resolution, it is possible to decrease the background DCR while maintaining a high analysis efficiency, as discussed in Section 3.1.2.

Superconducting Nanowire Single-Photon Detector

Similar to TESs, Superconducting Nanowire Single-Photon Detectors (SNSPDs) are a narrow superconducting thin film grown on a substrate that is patterned in various wire-like structures to limit the volume of the superconductor while still covering a large surface area for efficient photon absorption [29]. The wire is then cooled below the superconducting transition temperature, forming Cooper pairs, and biased with a constant current. When a photon is absorbed in the superconducting lattice, it deposits the energy by breaking Cooper pairs, resulting in a local non-superconducting region, commonly referred to as a hotspot [30, 31]. As the hotspot grows to span the entire wire, the system undergoes a swift and pronounced rise in resistance, measurable via the readout circuit. After the excess current has propagated through the readout circuit, the nanowire returns to the superconducting state [30, 31, 32]. The SNSPD are generally not energy-resolving devices due to the low energy threshold of their response. However, they have only nano-second scale dead times while maintaining similar quantum efficiencies as TESs [21, 33, 34].

3.1.1 Blackbody radiation

Blackbody radiation is the thermal energy radiated from a body in the form of electromagnetic waves (light). The spectrum of emitted light from a blackbody is described by Planck's law [35], and in terms of wavelength is:

$$B(\lambda, T) = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1} \qquad (3.1.3)$$

where k_B is the Boltzmann constant, h is the Planck constant and c is the speed of light. From Figure 3.1.2 it is clear that most of the blackbody radiation at 300 K is well within the mid and far infrared region, but the tail of the spectrum results in a non-negligible radiation at laser wavelength $\lambda = 1064$nm used in the ALPS II experiment. It is this effect which causes the majority of the increase in extrinsic DCR in both the TES and SNSPD [21, 36, 37].

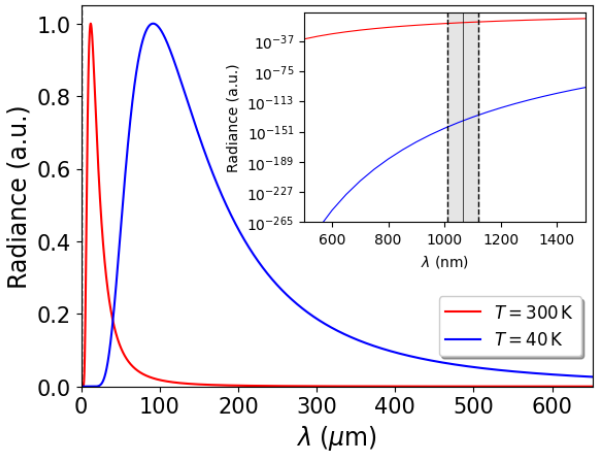


Figure 3.1.2: Blackbody radiation for $T = 300$ K and $T = 40$ K. Inlet shows the blackbody radiation for Red-NIR wavelength with ±55 nm TES energy resolution marked as a grey area.

3.1.2 Detection Significance

Detection significance is a statistical measure used in experimental science to quantify the confidence that an observed signal or effect is not due to random chance, but rather the result of a real underlying phenomenon. Detection significance is closely related to statistical significance, which is a measure of hypothesis testing. Consider two hypotheses, the null hypothesis H_0 and alternative hypothesis H_1, and a measurement whose value is a test statistic X, that is represented by the probability density functions $f_0(X)$ and $f_1(X)$, respectively. By comparing these two distributions, one can assess how well an observed value x_0 supports either hypothesis. The overlapping area of the two PDFs represents the probability that x_0 could plausibly arise under either hypothesis. The smaller the overlap between these two hypotheses, the greater the confidence that the measurement favours one hypothesis over the other [38, 39].

The detection significance for a photon counting experiment is determined by the number of signal photons (N_s) and the number of background events (N_b). In real experiments, the probability density functions of measuring n events, is described by the Poisson distribution [40],

$$f(n, \langle n \rangle) = \frac{\langle n \rangle^n}{n!} \mathrm{e}^{-\langle n \rangle} \tag{3.1.4}$$

where $\langle n \rangle$ is the average number of events. Note that for $\langle n \rangle \gg 1$, the Poisson distribution approaches the Gaussian distribution, as an example see Eq. 3.1.1. The two hypothesis for the axion search is: axions do not exists $\langle n \rangle = \langle N_b \rangle$, or axions do exist $\langle n \rangle = \langle N_b \rangle + \langle N_s \rangle$ [41, 39]. The overlapping area of the two PDFs (κ) is numerically shown to be [41]

$$\kappa = 1 - \mathrm{erf}\left(\frac{\sqrt{N_b + N_s} - \sqrt{N_b}}{\sqrt{2}} \right) \tag{3.1.5}$$

where $\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x \mathrm{e}^{-t^2} dt$ is the Gauss error function. The detection significance is then given by the variables in the argument of the Gauss error function [41]

$$S = \sqrt{N_b + N_s} - \sqrt{N_b} \tag{3.1.6}$$

and it is apparent from Eq. (3.1.5) that a larger S corresponds to a smaller overlapping area κ. Thus, a high value of S correspond to a lower probability that the data can be explained with the background only hypothesis. In the ALPS II experiment the detection significance is [42]

$$S = 2\sqrt{\tau}\left(\sqrt{\varepsilon_d \varepsilon_a n_s + n_b} - \sqrt{n_b} \right) \tag{3.1.7}$$

where τ is the measurement period, ε_a is the analysis efficiency probability of identifying a signal after detection, ε_d is the detection efficiency, n_s is the expected signal rate from theoretical considerations for photon-axion coupling and n_b is the background rate from extrinsic and intrinsic dark counts.

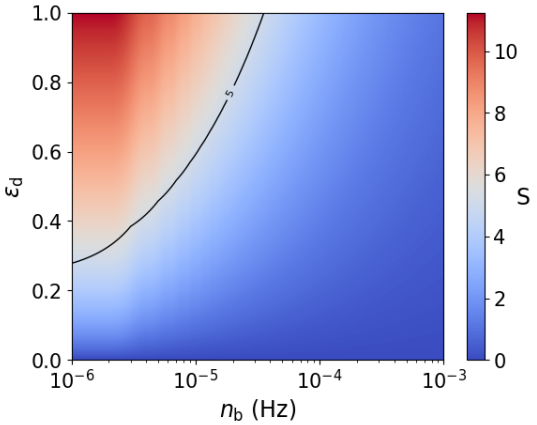


Figure 3.1.3: Contour plot of Eq. (3.1.7) with the theoretically estimated photon rate of $n_s = 1\,\text{photon/day}$ over a measurement period of $\tau = 21.5\,\text{days}$, as evaluated for the ALPS II experiment [19]. The analysis efficiency is set at $\varepsilon_a = 0.9$. The contour line illustrates the minimum requirement $S = 5$.

In general, for (astro-) particle physics experiments, $S = 5$ is agreed upon as a minimum requirement for an observation to be labelled as a new discovery [38, 41]. The detection significance S is shown in Figure 3.1.3 as a function of background rate, wit $S = 5$ shown as a solid line. Reducing n_b is the most effective way to increase S, apparent in Figure 3.1.3. However, since n_b and ε_d are closely related, this always comes at the cost of losing detection efficiency. Since the dependence of S from ε_d is less severe, reducing the n_b to the range of $10^{-6}\,\text{Hz}$ at a trade-off for the detection efficiency to drop to ~ 0.5 is permissible and will yield the required $S = 5$. This motivates the implementation of the COFB, as it improves the effective energy resolution of the TES resulting in an immense reduction of n_b while maintaining above 50% transmission, already proved possible by the initial tests performed in this thesis.

3.2 Cryogenic optical filter benches

As concluded in the previous section, for the ALPS II experiment to reach statistically significant results on the existence of axions the extrinsic DCR from blackbody radiation must decrease significantly. One method to achieve this is to use an optical filter cooled down to low temperatures, a method that has already proved significantly beneficial for SNSPD measurements [43, 44, 45, 46]. The optical filter used in the system is an ultra-narrow band-pass filter, see details in Section 3.2.2, with a transmission window of $\pm 1\,\text{nm}$. The practical implementation of this filter requires one to construct an opti-

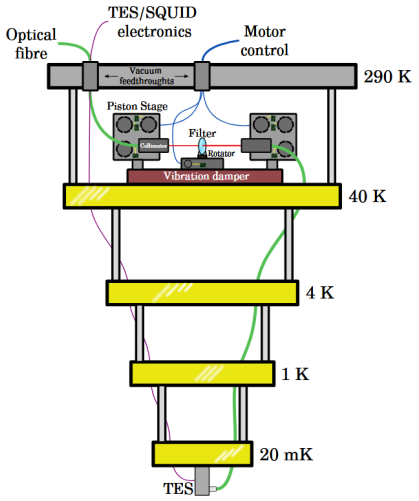


Figure 3.2.1: A schematic illustration of the planned integration of the COFB inside the dilution refrigerator for the ALPS II experiment

cal U-bench that can be operated at cryogenic tempera-
tures, see Figure 3.2.1. This comes with two major challenges: i) the filter bench must be placed in an isolated vacuum environment when cooled to below 0° to prevent moisture condensation. ii) The thermal contraction of the components after the cooldown changes the transmission window of the narrow band-pass filter and results in misalignment of the whole U-bench manifested as high optical losses. This misalignment cannot be corrected manually since the U-bench is in an isolated vacuum environment.

To resolve this, remotely controllable cryo-compatible piston stages are used. Additionally, to fine-tune the transmission window of the narrowband optical filter, the system incorporates a cryo-compatible rotator. Software to efficiently operate the cryogenic optical filter bench (COFB) needs development. The principal figure in the development of the COFB, is the transmission coefficient:

$$\mathrm{Tc} = \frac{P_{\mathrm{transmission}}}{P_{\mathrm{reference}}} \tag{3.2.1}$$

which directly correlates with the SDE of the single-photon detector.
Achieving high transmission requires efficient;

1. Collimation of light from/to emitting/receiving optical fibre
2. Alignment of collimated beam output/input
3. Identification of the transmission maximum over the parameter space.

Additionally, to maintain a high level of alignment this process should ideally be automated, to avoid stability issues during long measurements. This can prove an unusual challenge as motor hysteresis, arising from repeated fine adjustments or direction changes, may adversely impact system stability, especially as manual intervention is impractical during long measurements.

3.2.1 Thermal expansion

Thermal expansion is a consequence of anharmonic terms in the potential energy of the mean separation of a pair of atoms at a temperature T [47]. Thermal expansion for a solid is independent of the pressure in the system, and the volumetric thermal expansion coefficient is given as

$$\alpha_V = \frac{1}{V}\frac{dV}{dT} \tag{3.2.2}$$

where V is the volume of the solid and dV/dT is the change in volume per temperature change. For isotropic materials, the volumetric coefficient is proportionally related to the linear and area coefficients as

$$\alpha_V = 3\alpha_L = \frac{3}{2}\alpha_A \tag{3.2.3}$$

Thermal expansion coefficients in solids can be highly temperature dependent at cryogenic temperaturesThe COFB is currently assembled onto an aluminium body with $\alpha_L^{(\mathrm{Al})} = 23 \cdot 10^{-6}\,\mathrm{K}^{-1}$ [48], and will experience thermal contraction when cooling down from room temperature to the 40 K stage, see Figure 3.2.1. Furthermore, the collimating lens cavity is made of titanium, which exhibits a linear thermal expansion coefficient of $\alpha_L^{(\mathrm{Ti})} = 8.41 \cdot 10^{-1}\,\mathrm{K}^{-1}$ [49]. Titanium is chosen to match the thermal expansion coefficient of the collimating lens, which for borosilicate classes have thermal expansion coefficients of $3 \cdot 10^{-6}\,\mathrm{K}^{-1}$ [48]†. Thermal expansion coefficients presented in this work were determined at room temperature. If the thermal expansion is considered constant the change in length L of an object when cooled down is

$$L_{\text{final}} = L_{\text{initial}}[1 + \alpha_L(T_{\text{final}} - T_{\text{initial}})] \tag{3.2.4}$$

3.2.2 System components

This section presents the theoretical principles of optical components used in the design and measurement setup of the COFB, as well as relevant aspects of the specific components used in the Dark Quantum Lab (DQL) at the University of Southern Denmark (SDU).

Optical fibres

An optical fibre is a flexible solid cylinder made from a transparent dielectric material with a higher refractive index than the surrounding medium, such that light waves directed into the cylinder undergo a long string of internal reflections and are emitted on the other side with negligible losses. Inside the optical fibre, the rays are refracted according to Snell's law [50]

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2) \tag{3.2.5}$$

but since $n_1 > n_2$ it is possible for the system to experience no refraction, namely at angles above θ_{crit} (critical angle) which is derived from (3.2.5) and geometric considerations to be [50]

$$\theta_{\text{crit}} = \sin^{-1}\left(\frac{n_2}{n_1}\right) \tag{3.2.6}$$

The encasing medium (n_2 in figure 3.2.2) is known as the "cladding".

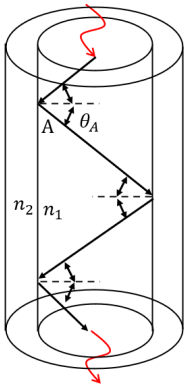


Figure 3.2.2: Optical fibre sketch

†The glass material used by Schäfter + Kirchoff is not publicly available.

If a ray encounters the cladding at a lower angle refraction can occur and the reflected ray will have lower energy. As the ray moves through the optical fibre it will eventually have lost most of the initial energy, due to the typical amount of reflections in such a fibre. A ray encountering the cladding at an angle higher than the critical angle will only be reflected and not lose energy due to propagating along the fibre (such as beam A in figure 3.2.2), only if the optical fibre is bent can non-reflecting angles occur[‡] [50, 51].

Single-mode optical fibres

Light can propagate in optical fibre in different modes, the number of which is proportional to the diameter of the fibre core. In single-mode fibres, the core diameter is narrowed so much that only single mode can propagate through. The amount of modes that can exist in an optical fibre is given by [50]

$$m_{\max} = \frac{1}{2}\left(\frac{\pi d}{\lambda}\sqrt{n_1^2 - n_2^2}\right)^2 \tag{3.2.7}$$

where the numerical aperture is $\mathrm{NA} = \sqrt{n_1^2 - n_2^2} = \sin(\theta_0)$ with θ_0 the maximum angle of incidence for internal reflection [48], d is the core diameter of the optical fibre, λ is the wavelength of the incoming light, and $m_{\max}$ is the maximum integer mode allowed for constructive interference to occur. This can also be expressed as a condition for single-mode behaviour

$$\frac{d}{\lambda} < \frac{2}{\pi(\mathrm{NA})} \tag{3.2.8}$$

which directly relates the core diameter and the wavelength of the propagating light[§].

Optical fibres with $m_{\max} \geq 2$ are known as multi-mode fibres while optical fibres with $m_{\max} < 2$ are called "single-mode fibres" and are the primary interest of this thesis. One of the physical aspects of single-mode optical fibres is their Gaussian beam profile [48], which in the context of this thesis, is used to validate the COFB ability to align towards the global maximum.

In the context of the ALPS II project, there are several reasons for using a single-mode fibre: Since the current TES detectors have dimensions of 25 micrometers [28], a larger diameter core emits the photons in a larger surface area resulting in poor detection efficiency, single-mode optical fibres usually have core width in the ∼10 micrometre range, while multimode fibres ∼50 micrometres. Furthermore, the choice of single-mode fibres is motivated by significantly reduced coupling of blackbody photons [46].

[‡]A more comprehensive understanding of optical fibres requires an in-depth treatment of optical waveguides and Maxwell's equations with appropriate boundary conditions. However, such analysis lies beyond the scope of this thesis.

[§]An in-depth analysis indicates that $2 \to 2.4$ also results in single-mode effects.

In the measurement setup for the COFB THORLABS P3-1064Y-FC-1 single-mode optical fibre is used, which exhibits an attenuation of 1.5 dB/km (see next section), mode field diameter (MFD) of $6.2 \pm 0.4\,\mu$m and NA of 0.14 [52].

Attenuators

Attenuation is the measure of power loss light experiences when propagating through a fibre and is measured in decibels (dB) per unit length (usually kilometres) as

$$\alpha(\text{dB}) = 10 \log_{10}\left(\frac{I_{\text{in}}}{I_{\text{out}}}\right) \tag{3.2.9}$$

where I_{in} and I_{out} are the intensity cross-section of the fibre at the two ends.
Attenuation is subdivided into two categories: extrinsic and intrinsic.

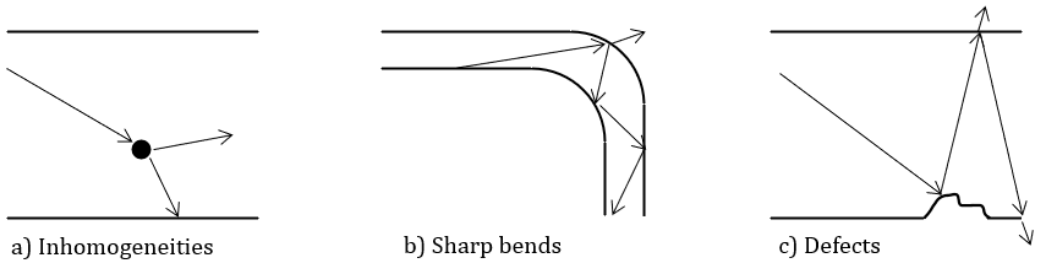


Figure 3.2.3: Different types of optical fibre attenuation

Intrinsic attenuation (Figure 3.2.9, a) is due to the properties of the optical material used in the fibre, impurities can lead to internal absorption and microscopic inhomogeneities can lead to Rayleigh scattering. Extrinsic attenuation (Figure 3.2.9, b, c) is caused by geometric effects from defects, sharp bends and coupling losses from mating sleeves [51].
The COFB testing setup incorporates a THORLABS VOA1064-APC variable attenuator, which operates on a similar collimating-decollimating lens connection as the COFB, but with a blocking device instead of a filter, which is moved linearly into the beam-path by a screw mechanism. The variable attenuator has an intrinsic attenuation of 1.5 dB when not engaged [53].

Collimation of Gaussian beams

The collimating lenses collect emitted radiation from an optical fibre and align the light rays in parallel to minimize propagation spread. This is achieved by considering the emitting optical fibre as a point source and aligning the collimating lens such that this point is at the focal point

of the lens [48]. For spherical Gaussian beams, the thin lens formula is [54]

$$\frac{1}{f} = \frac{1}{s} + \frac{1}{s' + \frac{z_R'^2}{(s'-f)}} \tag{3.2.10}$$

where f is the focal length of the lens, s is the object distance, s' is the image distance and z_R' is the Rayleigh range of the image [48]. The Rayleigh length is defined as

$$z_R = \frac{\pi w_0^2}{\lambda} \tag{3.2.11}$$

where w_0 is the beam waist radius and λ is the wavelength of the incoming light. The Rayleigh length is the distance from the beam waist, the narrowest width of the beam, to the position where the cross-section is doubled. The image and object Rayleigh length are related as [54]

$$z_R' = m^2 z_R \tag{3.2.12}$$

where $m = 1/\sqrt{[1-(s/f)^2)]^2 + (z_R/f)^2}$ is the magnification of the lens. In the case of a single-mode optical fibre coupled to a collimating lens, the MFD of the fibre can be understood as twice the beam waist radius of an incoming Gaussian beam, see Figure 3.2.4

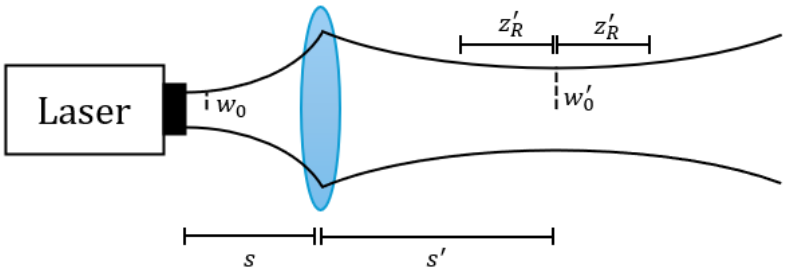


Figure 3.2.4: A schematic illustration of collimation of a Gaussian beam with relevant length scales.

From λ, w_0, s and f, the image distance s', Rayleigh length z_R' and beam waist radius w_0' of the outgoing beam can be determined analytically. As the object distance s approaches the focal length f, the image distance goes towards infinity resulting in a perfectly collimated lens, as the second term in Eq. (3.2.10) must disappear.

The COFB utilizes a set of Schäfter+Kirchoff 60FC-4-A8-07 fibre collimating lenses with a working range of 650nm to 1150 nm, coupled through an FC-APC connection and with a focal length of 8 mm for $\lambda = 1064$ nm. By rotating the collimating lens it acts as an in-coupler,

collecting the collimated beam and returning it to a source-like point, where it can be re-captured into an optical fibre. The lens z-position (along the focal axis) can be adjusted by an eccentric key in the outer casing of the lens and locked by two clamping screws [55].

FC-APC connections and PM mating sleeves

The optical fibres are in contact throughout the system using the FC-APC connectors. These connectors have low transmission attenuation but high reflection attenuation, due to an 8° angle which causes reflected rays from the core-coupling to be directed into the cladding. Polarization-maintaining (PM) mating sleeves are designed so only one state of polarization is transmitted through the connections.
In the COFB the optical fibres are connected using THORLABS ADAFCPM2 - FC/APC mating sleeves, which exhibit low connection attenuation of less than 1.0 dB [56].

Beamsplitter

A beamsplitter is an optical device that separates an incoming beam of light into two beams, a reflected and transmitted beam, or vice versa. Beamsplitters are achieved by fusing two optically transparent materials, with another refractive material in between creating a boundary for the incoming wave to reflect/transmit on/through [48]. By adjusting the thickness/type of this material it is possible to achieve a wide variety of reflection/transmission ratios.
The COFB incorporates two THORLABS Narrowband ($1064\text{nm} \pm 15\,\text{nm}$) fibre optic couplers, with 50%:50% split ratio and attenuation loss smaller than 3.5 dB [57].

Photodiode power meters

Photodiodes are semiconductor devices based on a p-n junction, which creates an energy band gap that allows the device to convert incident photons into an electrical current. The best photodiode detectability is achieved by operating it in reverse-bias, where a positive voltage is applied to the n-type side of the junction. In this configuration an incident photon induces a change in current through a load resistor, yielding a linear response to changes in light intensity [48].
In the measurement setup for the COFB two THORLABS PM16-144, integrating sphere sensor, power meters are used to evaluate the reference and transmitted power of the laser source. The PM16-144 has a working optical power range of $1\,\mu\text{W} - 500\,\text{mW}$ with a linearity precision of $\pm 0.5\%$ (for a beam diameter of less than 1 mm) and a power resolution of less than 1 nW [58].

Fabry-Perot Laser source

The Fabry-Perot laser source is a type of laser diode which utilizes a resonance cavity to achieve wavelength stability [59]. Laser diodes produce coherent light through stimulated emission, which is emitted into an optical structure consisting of two reflecting mirrors positioned such that only photons at a specific frequency are amplified and afterwards emitted. Transmission through the reflecting mirrors is governed by the condition [60]

$$2nL\cos\theta = m\lambda \tag{3.2.13}$$

where n is the refractive index of the medium inside the cavity, L is the length of the cavity, θ is the angle of incidence, m is the mode number and λ is the wavelength of the transmitted light. The THORLABS S1FC1060 benchtop laser source is used in the testing setup for the COFB. The S1FC1060 is a class 3B laser source, see section 3.2.3 for more detail, emits light at $\lambda = 1064\,\text{nm}$, has min full output power 20.0 mW and a stability of ± 0.1 dB after operating for 24 hr [61].

Piston stage / Piezo step-motor

A Piezo motor is an actuator that relies on the piezoelectric effect to generate mechanical motion, linearly or rotary [48]. The piezoelectric effect is the physical ability of certain materials, ceramic or biological matter to accumulate electric charge when experiencing mechanical stress. Piezo motors utilized the reverse piezo effect (applied electric field induces mechanical strain) of these materials, in the case of the COFB, to generate linear motion.
In the COFB the CTTPS1 is equipped with three piezo step-motors/Cryo Linear Actuators (CLA), allowing tip/tilt motion. CTTPS1 utilize three CLA2201 piezo motors, which yields a total tip/tilt range of $\pm 5.6°$, resulting from a total piston range of $\pm 1.5\,\text{mm}$, with a minimum step size $\sim 1\,\text{nm}/1\,\mu\text{rad}$. The piston-stage and piezo-motors are operational from 4 K to 375 K, with some system specification (such as motor velocity and minimal step size) temperature-dependent [62, 63].

The motors are self-locking to avoid power dissipation when not operating. The tip/tilt mounting part is connected through preloaded springs to avoid any subtle motion. The system is controlled using Python and connected through two JPE control modules, as shown in Figure 3.2.5. In this text, references to "motor #" corresponds to the cable numbers shown in Figure 3.2.5.

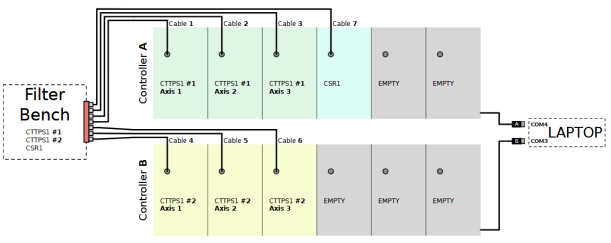


Figure 3.2.5: Connection diagram

Optical filters

Optical filters only transmit a specific range of wavelengths, while providing high attenuation for the rest of the spectrum. The attenuation level of an optical filter is measured in optical density (OD) which is a log-10 measure of the absorbance of the given laser wavelength [48]:

$$\mathrm{OD} = -\log_{10}(T) \tag{3.2.14}$$

where T is the transmission. Optical filters are characterised by the centre wavelength and FWHM, which can be modified by tilting the filter such that the angle of incidence (AOI) changes, effectively blue-shifting the transmission spectrum. The central optical filter for the implementation of which the COFB is built is the Alluxa 1065-1.8 OD6 Ultra Narrow Bandpass Filter, which has a central wavelength of 1065 nm and a FWHM of less than 2 nm. This will be installed on a rotator in the centre of the COFB to blueshift the spectrum to reach the desired 1064 nm transmission wavelength. The filter has a OD6 blocking level in the 800 nm − 1051 nm and 1079 nm − 1300 nm ranges, translating to a transmission of 0.0001% [64].

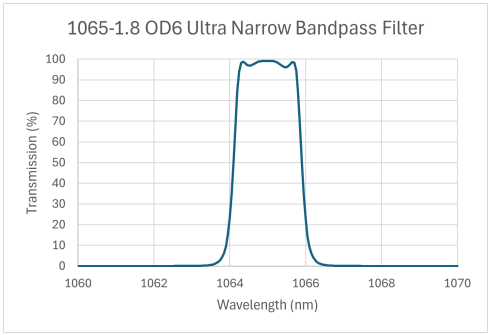


Figure 3.2.6: Filter transmission band (data from [64])

Alignment tools

The initial collimating phase of the alignment process requires a manual qualitative evaluation of a beam profile, by visually evaluating how the size of the collimating beam varies as a function of distance.

Detector card: Since the system is designed to operate at near-IR wavelengths, a medium to convert the 1064 nm light into a visible wavelength is needed. The THORLABS VRC2 - VIS/NIR Detector Card [65] utilized slow-fading phosphor, which exhibit luminescence emission at $\sim$ 580 nm − 750 nm from absorbing light in the 400 nm − 640 nm and 800 nm − 1700 nm bands. The card is credit-card sized and has an active photosensitive region of 53.3 mm × 30.5 mm.

Handheld laser: Due to potential initial misalignment of the piston stages, there exists a significant risk that the outgoing laser beam fails to pass through the aperture in the receiving piston stage, instead impinging on the structural surface and reflecting backwards. This reflection would be hard to detect before activating the 1064 nm laser resulting in unnecessary safety risks from reflections, and additionally reducing the operational efficiency of the collimating phase. To

avoid these challenges, before adjusting the collimating lens to the near-IR light an additional adjusting step is included using a visible wavelength laser source. The IMM Fiberpoint® 250 [66] handheld laser source is a class 2 laser, emits visible light with wavelength $\lambda = 635\,\text{nm}$ and has a maximal optical output power of less than 1 mW.

3.2.3 Safety

Exposure to laser radiation can result in permanent damage to eyes and skin, especially the retina [67]. Laser damage can stem from several mechanisms, such as thermal, photochemical or photoionization. The possible injuries vary with wavelength and for the context of this thesis, and the usage of a laser-source emitting 1064nm wavelength radiation, the main cause for concern is retinal burn and the formation of cloudy areas in the eyes known as cataracts. The maximum permissible exposure (MPE) is the highest power of a light source with negligible probability of causing damage and is measured in W/cm^2. Laser sources are categorised into classes to standardize the necessary safety precautions.

The THORLABS S1FC1060 laser source is classified as a Class 3B laser and is highly hazardous if exposed directly to the eye, but diffuse reflections (from matte surfaces) are not harmful. To avoid eye damage the DQL is equipped with Lasershield protective eyewear, which is graded OD6+ for $910\,\text{nm} - 1070\,\text{nm}$ wavelengths.

The IMM Fiberpoint® 250 is classified as a Class 2 laser and is generally considered safe for operation, provided that direct eye exposure to the beam is avoided.

Chapter 4

Experimental Implementation of the Cryogenic Optical Filter Bench

4.1 System Overview and Schematics

The experimental setup is assembled onto a Newport Research Corporation optical table. The picture in Figure 4.1.1 was taken at the end of the project duration.

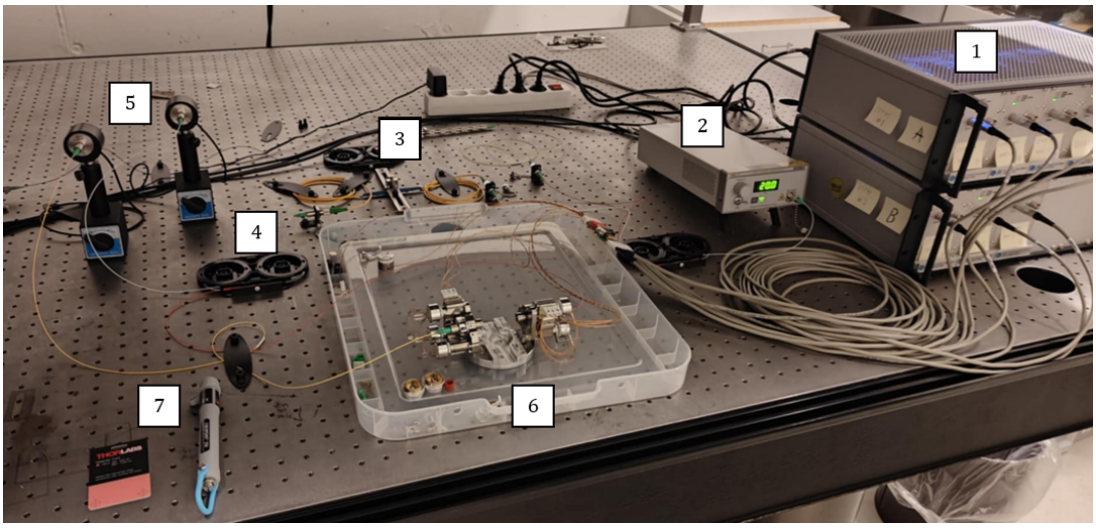


Figure 4.1.1: Dark Quantum Lab (DQL) optical measurement setup

List of components as seen in Figure 4.1.1:

1. Piston stage control (3.2.2)
2. Thorlabs 1064 nm Laser Source: [61]
3. Thorlabs Variable Attenuator: [53]
4. Thorlabs Beamsplitter: [57]
5. Thorlabs Power Meters: [58]
6. Cold optical filter bench (COFB) (4.1.2)

7l. Thorlabs Detector card: [65]

7r. IMM photonics Fiberpoint 250, red laser: [66]

4.1.1 Measurement setup schematics

The Optical filter bench is connected to the measurement/testing setup as follows

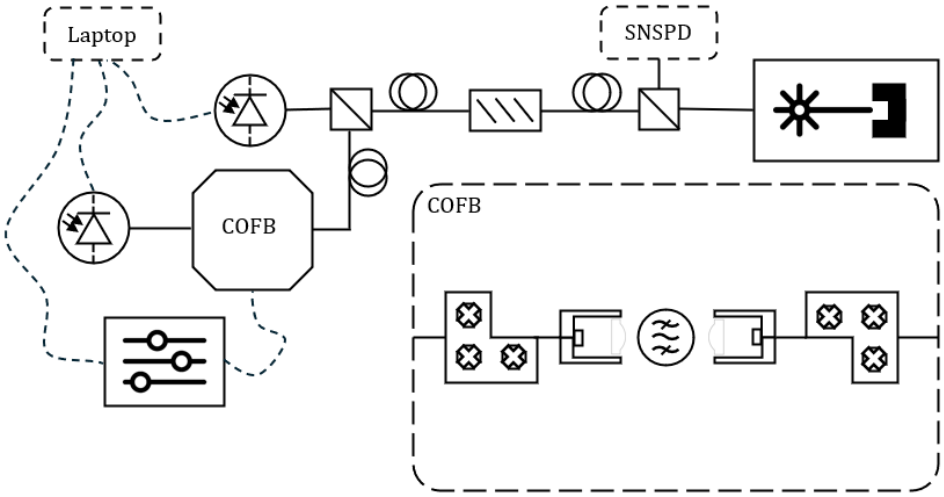


Figure 4.1.2: COFB testing setup. The first beamsplitter and variable attenuator will not affect the experimental measurements of the COFB.

The laser source injects a beam of 1064 nm light which is subsequently split 50%:50% through a beamsplitter, with one output going into the testing system and the second output into a secondary testing system (in figure 4.1.2 testing of an SNSPD). The beam then goes through a variable attenuator and into a second 50%:50% beamsplitter, where one output goes to the reference power meter and the second goes into the COFB. In the COFB the beam is emitted and collimated in an adjustable collimating lens, passed through a narrowband optical filter, and decollimated in a second adjustable collimating lens. The collimating/emitting and decollimating/receiving lenses are spatially controllable through two piston stages that are operated through a set of control modules, Figure 3.2.5. The beam is then finally transmitted into the transmitted power meter. The power meters are connected to the readout computer and are controlled in the same Python environment as the piston stages.

Symbol	Description
	Laser source
	Beamsplitter
	Variable attenuator
	Power meter / photodiode
	Piston control system
	Optical band filter
	Tip/tilt - Piston stage
	Adjustable Collimating Lens
	Electrical wiring
	Optical fibre

4.1.2 Cryogenic optical filter bench in detail

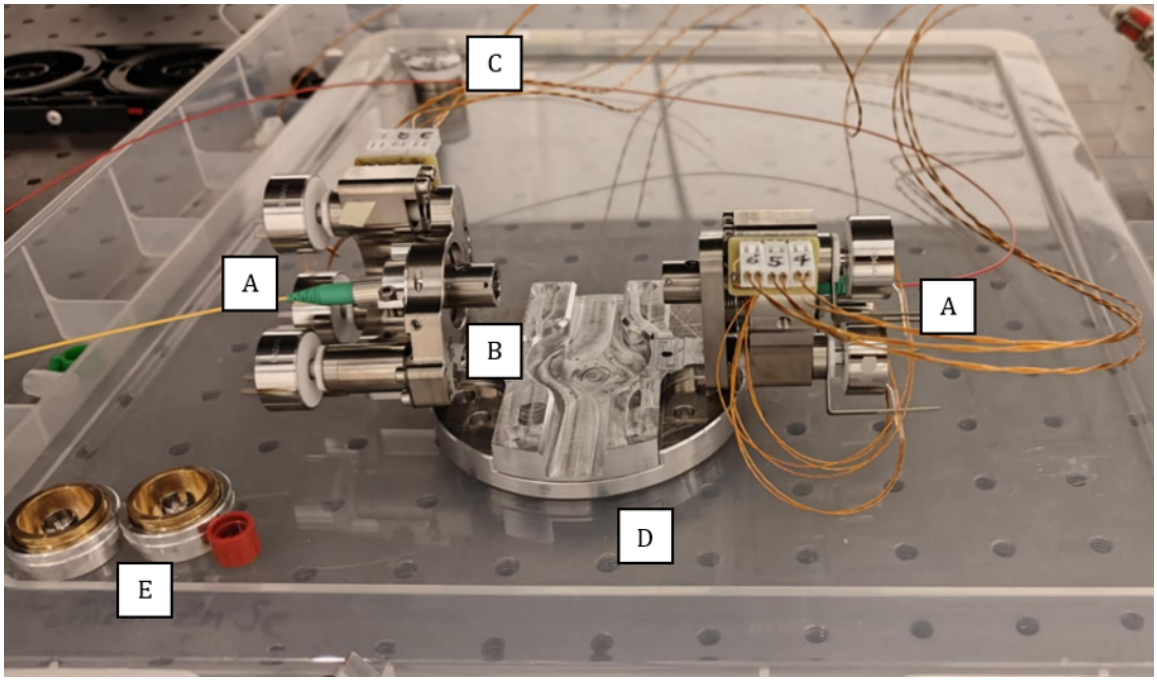


Figure 4.1.3: Photograph of the Cold optical filter bench, taken at the end of the project.

The COFB design, Figure 4.1.3: (A) two CTTPS1 - tip/tilt/piston stages [63], each equipped with three CLA [62], (B) two collimating lenses [55], (C) a CRS - cryo sample rotator [68] that was not utilized in the experiments involved with this theses, (D) an aluminium base plate designed at SDU, to be later attached to the vibration damper (not installed on 4.1.3) and an optical filter [64] which was not used in the experiments involved with this thesis. On the bottom left (E), the fibre-couplers for the next installation of the COFB are shown.

4.2 Alignment methodology

This section presents the alignment procedures and operation of the COFB, performed before installing the optical filter. The two types of interaction with the system are classified in this thesis as *manual adjustment* and *digital control/measurement* referring to the physical manipulation of the setup and scripted control using Python respectively. These types of interaction refer to post-assembly where no significant changes are made to COFB or surrounding measurement system*.

*During the project period the optical system was updated to incorporate a second output beam, referring to the first beamsplitter from the laser source in fig 4.1.2, and a variable attenuator, to avoid safety issues when using the secondary output beam, as well as slight modification to the power-meters to avoid temporary sensor failure. These changes will not affect the transmission coefficient as large modifications precede the second beamsplitter, which couples to the COFB and the two power meters.

Manual adjustment

The system allows for manual adjustment through two mechanisms: rotating the variable optical attenuator within the measurement setup to block the beam and employing an eccentric key, or narrow flat-headed screwdriver, to modify the position of the collimating lens within the connection cavity, as show in Figure 4.2.1.
Furthermore, it should be noted that during the reconnection of optical fibres to various components of the setup, the fibre end faces are cleaned using a spool of lint-free cleaning cloth moistened with a small amount of ethanol, following standard practices.

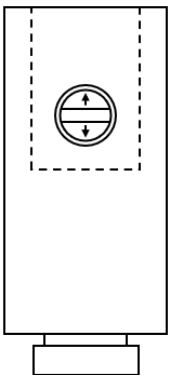


Figure 4.2.1: Collimating lens sketch.

Digital control and measurement

The COFB is controlled via a Python environment and, for the duration of this thesis, operated within Visual Studio Code as a Jupyter Notebook.

The JPE actuators on the piston stages are operated as stepping motors, within a safety range of $\pm 80,000$ steps set by the operators. The motors possess limited onboard memory while powered, and upon activation, they initialize their position at step value zero, which serves as the reference point. The primary method of motor control employed during this thesis involves inputting a target integer position, where positive values correspond to clockwise rotation (inward movement relative to the filter position of the COFB), and negative values indicate the opposite. Operating the motors also requires a user to input the step frequency (movement speed) between $1-600\,\mathrm{Hz}$ and temperature of the environment (in Kelvins)[†]. In addition, the motors have limited internal safety measures, which can be modified to define the allowed integer step range the motor can take. Furthermore, an internal measure subdivides step sizes above $50,000$ to be done over multiple executions. The zeroth step value can be set dynamically during operation.

The used power meters require a total measurement time in seconds, sampling interval in seconds and an operating wavelength, and produce a list of measured power values in mW along with corresponding timestamps set through user input. Throughout this thesis, each measurement of transmission and reference power was averaged for $0.75\,\mathrm{s}$, with a sampling interval of $0.05\,\mathrm{s}$. The corresponding standard deviation for each measurement was also recorded.

[†]For the duration of the thesis the temperature was set to $293\,\mathrm{K}$.

4.2.1 Detailed alignment procedure

The alignment procedure consists of four sequential phases. A trial-and-error investigation of this approach is presented in section 5.1; however, the procedure outlined below yielded the most consistent and reproducable results.

Preliminary setup phase

Before operating the filter bench with the 1064 nm laser source the alignment of the emitting piston stage (ePS) is initialized to ensure transmission to the receiving piston stage (rPS). This is achieved by replacing the collimating lens and connection cavity from the rPS, with a test aperture with a small hole located at the position where the receiving fibre would be attached. Afterwards, the 635 nm handheld test laser can be connected. The laser beam should subsequently be visible either on a plane behind the rPS or directly on the rPS surface. If the beam appears on the rPS itself, adjustment of the ePS via the tip/tilt motors is required to correct the alignment. After ensuring beam passage through the aperture, the ePS can be collimated[‡]. This collimation does not require extreme precision or excessive time, as the focal length of the collimating lens will vary when switching to the 1064 nm laser source. Additionally, the motors on both the ePS and rPS should be adjusted such that the tops of the mounting screws (light blue) lie coplanar with the backside of the aperture body (dark blue) and the piston stage housing (red), shown in Figure 4.2.2.

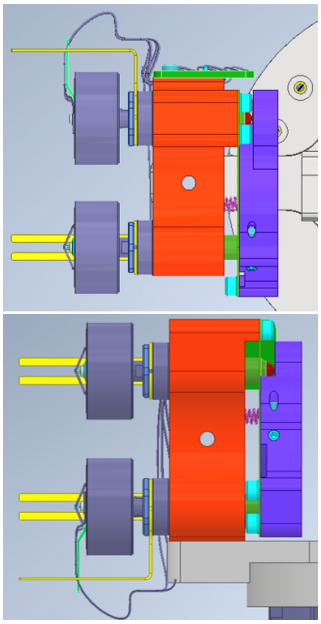


Figure 4.2.2: Digital piston stage model - top: top view, bottom: side view.

Initial collimating phase

Before re-inserting the collimating lens and connection cavity into the aperture of the rPS, the ePS lens must be adequately collimated. This process is particularly challenging to perform without assistance, as the detector card needs to be positioned at a considerable distance from the ePS to achieve precise collimation. In addition, by assuring that the beam diameter remains approximately constant as a function of distance, the collimation is preserved. This step is omitted for the visible laser, as both the focal length and, consequently, the degree of collimation are highly wavelength-dependent. After collimating the ePS, re-insert the collimating lens and connection cavity into the rPS. Roughly adjust the rPS collimating lens to have the same position

[‡]Throughout this text, the phrase 'emitting/receiving piston stage is collimated', or similar expressions, refers specifically to the alignment of the collimating lens connected to the respective piston stage.

as the lens in the ePS, by visually comparing the position of the manipulation line in Figure 4.2.1 (the line inside the innermost circle with two arrows pointing upwards and downwards). This method prevents alignment towards local maxima and internal reflections but entails the trade-off that the initial step of the alignment phase may require additional time. For the remainder of the procedure, the collimation of the ePS should be held constant, with efforts concentrated on the fine adjustment of rPS during alignment.

Primary alignment phase

The general scheme for aligning the piston stages follows an iterative method in which the motors are actuated one at a time from their current position (where *position* refers to the integer step value) to a negative starting point, i.e. first rotating the motor counter-clockwise to a point, and subsequently then traverses the positions to reach the endpoint, i.e. rotating the motor clockwise. This is done at a defined step size (also referred to as the step length) and stepping frequency (also referred to as movement speed). At each position, the transmitted power is measured. Upon finishing the scan the operating motor can be repositioned, typically to the position corresponding to the highest transmitted power, which is then defined as the new step-zero reference. After each motor iteration (i.e. each motor adjustment per piston stage cycle), the transmission coefficient is determined by measuring both the transmitted and reference powers. As an example, see Figure 4.2.3; a scan from position −100 to position +100 with a step length of 10 and a frequency of 10 Hz would move through positions −100, −90, −80, ..., 0, ..., +80, +90, +100, measuring the transmitted power at each position. In this example, one second is spent transitioning between each step position, and approximately three seconds are spent measuring the transmitted power at each position, with measurement duration defined in section 4.2. The highest transmitted power was measured to be at position +10, and the motors were subsequently moved to that position which was defined as the new step-zero reference.

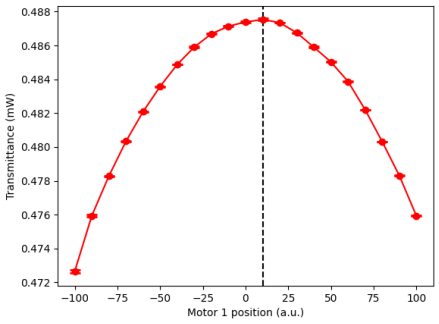


Figure 4.2.3: Fine alignment procedure of motor 1, from −100 to +100 with maximum transmitted power at +10, marked by the black dashed vertical line.

The alignment phase consists of the following steps:

(1) Commence by aligning the motors in the ePS and rPS iteratively in a position range of ±10k, with a step length 1k and frequency 600 Hz. Care should be taken to prevent the motors from disengaging or bottoming out. Continue this step until a prominent peak

appears on the transmission power curve. Using the visual alignment of the two collimating lenses inside the cavities, achieving optimal alignment generally required observing power peaks corresponding to transmission levels in the range of 10% to 20%. Then, by further iterative alignment, shift the peak towards the step-zero reference.

(2) Continue the alignment by decreasing the position range, step length and movement speed following the general ratio: 10:1:1[§]. Terminate the alignment once the peak is near the step-zero reference for all motors.

(3) Proceed to collimate the rPS using the real-time readout of the power meters[¶]. Collimating the rPS requires manual interaction with the system, and excessive force applied when inserting or operating the eccentric key can result in misalignment of the rPS.

(4) Complete the alignment process by repeating steps (2)-(3) for the rPS until further alignment yields negligible improvement in transmission.

In the absence of measurable transmission above the noise level, performing an initial alignment with a multi-mode optical fibre using the same procedure is often effective. Proceed by re-connecting the single-mode optical fibre and going through steps (2)-(4) starting with a position range of ±5k, step length 500 and movement speed 500Hz

Evaluation phase

To evaluate the alignment process, scan the ePS and rPS motors in the position range of ±1k. All six scans should yield a Gaussian transmitted power profile, accompanied by small symmetric side peaks caused by diffraction or internal reflections within the lens cavity.

The stability of the alignment process can be assessed by monitoring the transmission coefficient after each motor iteration. Significant fluctuations in the transmission coefficient may indicate instability in the alignment, see Figure 4.2.4

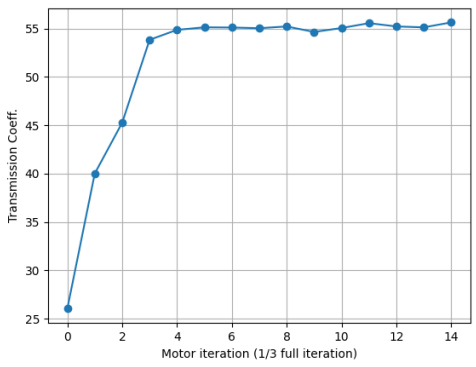


Figure 4.2.4: Transmission coefficient plot for rPS alignment procedure, from a set of scans over a ±1k position range.

[§]When a scan position range is mentioned in this text, the associated step length and movement speed follow this ratio, unless otherwise specified.

[¶]This operation is not part of the control script in appendix A due to its requirement for direct terminal access.

Chapter 5

Results and Optimization of the Cryogenic Optical Filter Bench Alignment Procedure

This chapter presents the experimental results obtained from the alignment procedure of the COFB outlined in Chapter 4. It details the optimization of the alignment procedure and outlines the strategies implemented to address the primary challenges of the system, such as internal reflection and procedure duration.

Control software development was the primary component of the optimization process, intended to introduce digital operability and support cryogenic deployment. The control software is provided in Appendix A and should be used in a Jupyter Notebook environment* for graph updates to work properly†.

5.1 Optimizing the alignment procedure

5.1.1 Locating the global maximum

The most significant challenge encountered during the procedure was accurately locating and maintaining alignment with the global maximum. Collimating the rPS can potentially result in aligning towards a local maximum, originating from internal reflections in the cavity or diffraction. Manual analysis of transmitted power graphs proved an acceptable method to avoid

*Visual Studio Code was used during the project.

†Following third-party libraries were utilized in the code (based on Python v3.10.11): numpy v1.26.4, matplotlib v3.8.3, IPython v8.22.2 and pynput v1.8.0

aligning towards non-global maxima. This was achieved by developing a non-automatic routine where a single motor scan would yield a transmitted power graph and subsequently, the user could input a position to set as the new step-zero position.

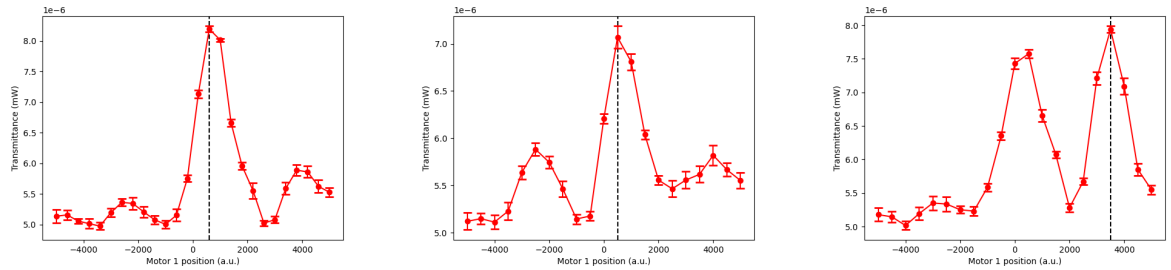


Figure 5.1.1: Transmitted power graphs for a rPS alignment procedure, from a set of scans over a ±5k position range.

In Figure 5.1.1 samples of transmitted power graphs are shown. The first high-transmission above 50% alignment was achieved by analysing a power graph similar to the rightmost figure in Figure 5.1.1, and manually relocating the scanning motor to the minima between the two peaks. At this time, the primary contributing factor to the presence of local maxima was believed to be diffraction. The underlying idea was to interpret each power graph as a cross-sectional line through a symmetric diffraction pattern, projected onto the rPS collimating lens. In Figure 5.1.2 circular diffraction patterns are presented with dashed lines representing scan paths. Top: illustration of the three types of motor scans. Middle: example scan displaying a double-peaked power graph corresponding to the left graph in Figure 5.1.1. Bottom: elongated profile indicative of a misalignment of the ePS, and showing a large peak at the centre of the scan. This visualization is however misguided as the scan is not a measure of power at a given position, but a measure of power at a given, albeit very tiny, angle. Multiple issues arise from this, as the tip motors have a rotational axis at the edge of the piston stage while the tilt-motor has a rotational axis through the centre of the collimating lens. A more thorough investigation of the impact of the different rotational axes is needed to make this method viable. During the project moving the tip motors in opposite directions was tried, to see if tilt-like rotation was feasible. This would reduce the complexity of analysing such power graphs, and reduce the duration of the maxima-finding phase of the alignment procedure. In practice, this proved cumbersome as motor hysteresis made control in different operating directions laborious and time-consuming while providing no significant in-

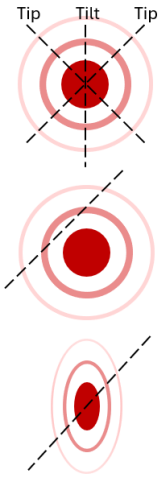


Figure 5.1.2: Circular diffraction patterns, with tip/tilt lines representing the scanning range for different motors classes.

crease in precision. For this method to have any relevant impact on the capability to identify the global maximum, a thorough investigation of motor hysteresis, both at room and cryogenic temperatures, is necessary. While this method was not pursued further during optimization, future integration of machine learning techniques to classify power graphs could offer significant advantages.

After noticing a clear tendency of the collimating lens in the rPS to be moved inwards during collimation steps, beyond the circular adjusting hole in figure 4.2.1, while achieving low transmission less than 10%, adjusting the lens to mimic the ePS initially proved sufficient for finding the global maximum after alignment and subsequently collimating the rPS. This method severely reduced the potential of aligning towards a non-global maximum and reliably produced high transmission above 50%.

5.1.2 Limiting accuracy loss and operational time for large scan ranges

Once a high degree of collimation accuracy was achieved, testing of alignment procedures became possible. Initially, the alignment followed a randomized motor sequence to explore the large six-dimensional parameter space, without guidance. This method yielded above 50% transmission but required extended durations and carried an increased risk of motor limit breaches. Consequently, alternative sequences for aligning the ePS and rPS were evaluated, with the initial alignment of the ePS followed by the rPS generally resulting in the fastest procedure.

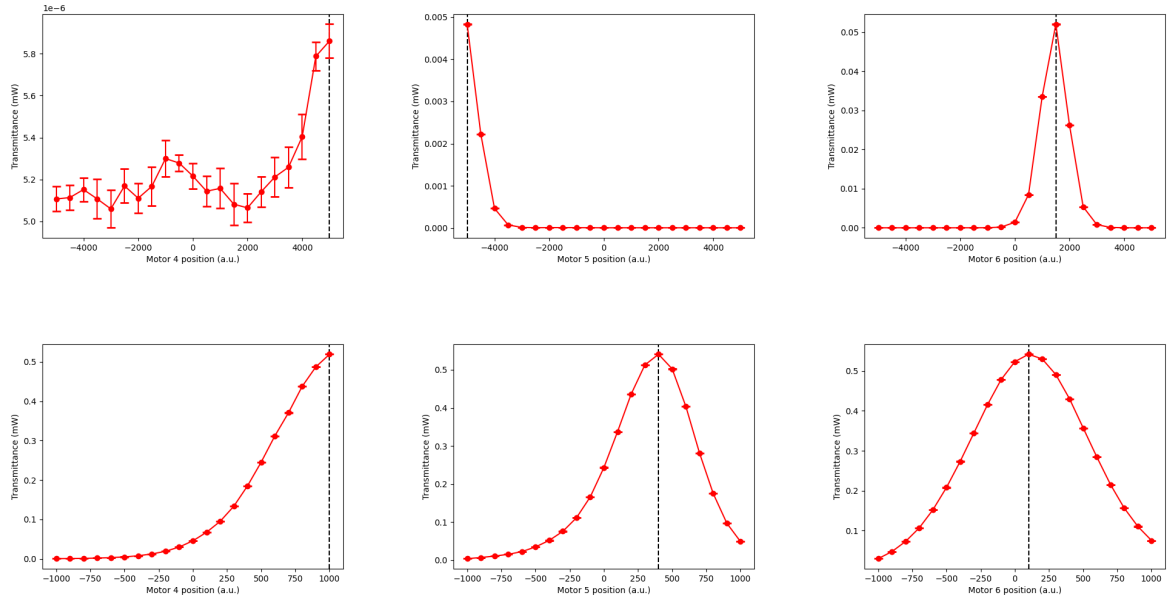


Figure 5.1.3: Examples of transmitted power graphs of a two-step ePS alignment procedure with: top row shows the scan over a ±5k position range; bottom row shows a follow-up scan over a ±1k position range.

As seen in figure 5.1.3, the transmission window of the ePS for the global maximum was ±1k steps, and similar ranges were also seen in the rPS, see Section 5.3. To avoid unnecessary operation duration a gradient cut-off subroutine, referred to as 'clever', was implemented. This subroutine checked if the transmitted power was increasing or decreasing after each iteration, and subsequently stopped the operation if the system was no longer increasing transmitted power. This subroutine reduced the average operation time of ±1k scans by $\sim$ 40% but had issues when aligning larger and shorter scan ranges. For long scan ranges, local maxima would stop the scans early, while for short scan ranges (around the global maximum) fluctuations in laser intensity had the same effect. Attempts to incorporate error considerations via a series of boundary protocols were unsuccessful. Manually operating the subroutine proved more efficient and considerable time was dedicated to implementing real-time keyboard control for interactive use during alignment. Additionally, several quality-of-life features were implemented during this process, such as the ability to pause and stop alignment operations.

The 'Clever' subroutine effect on alignment stability

Apart from reducing operation time, the 'clever' subroutine had a positive effect on alignment stability. Since the subroutine stopped the operation exactly one iteration past the global maximum only a single movement operation was required to align towards the maxima.

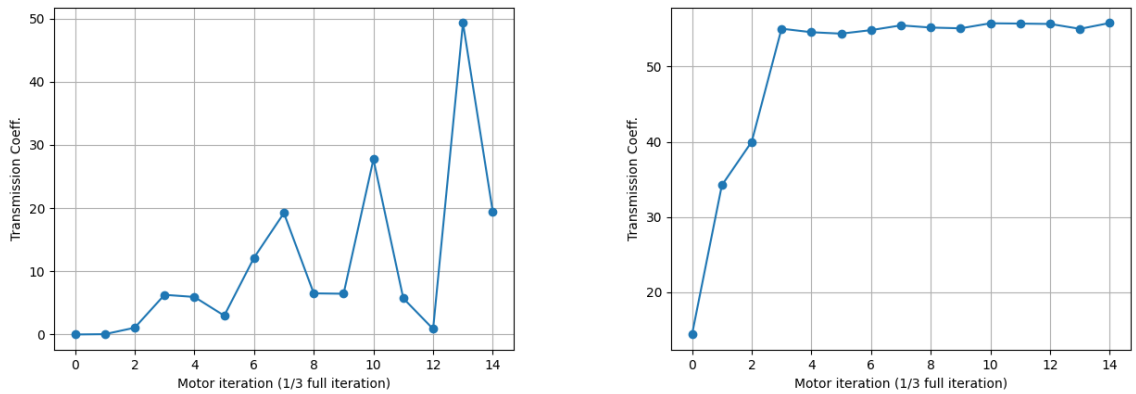


Figure 5.1.4: Examples of transmission graphs from a two-step rPS alignment procedure. Left; Transmission coefficients for a set of scans over a ±5k position range. Right; Transmission coefficients for a set of scans over a ±1k position range

As can be seen in figure 5.1.4, on the left since the scanning range was ±5k the 'clever' subroutine was not engaged, significant variations between consecutive iterations. On the right, the subroutine was activated and while fluctuations did occur after system alignment (past motor iteration 4), it emphasises the positive effect of the subroutine.

5.2 Challenges and limitations

5.2.1 Errors introduced by the optimized alignment procedure

The 'clever' subroutine significantly improved the alignment duration after collimation and alignment stability over long measurements. Unfortunately, introducing the subroutine and the accompanying keyboard control interface introduced minor error pathways into the alignment procedures.

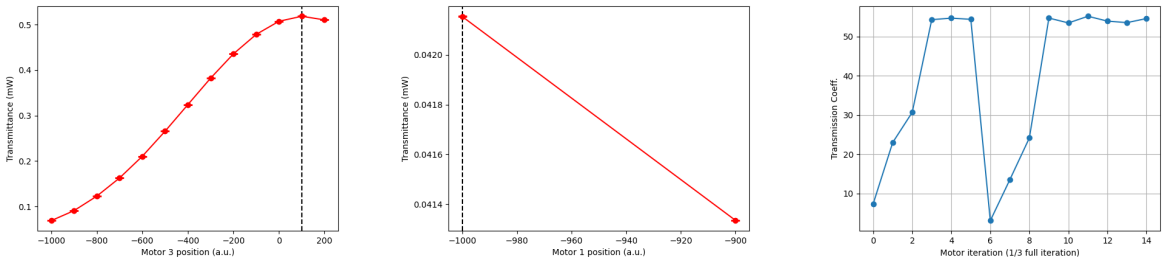


Figure 5.2.1: Left and centre: two consecutive alignment procedures. Right: transmission coefficient graph for a rPS alignment procedure, from a set of scans over a ±1k.

In figure 5.2.1 an example of the 'clever' subroutine localising a local maxima and stopping the alignment procedure caused a significant drop in transmission. The system subsequently realigned in the following three motor iterations, and the affected motor (iterations 6 and 9) managed to escape the local maxima trap.

Introducing the keyboard control inadvertently made non-COFB computer usage during measurement difficult, as certain keys activated/toggled functions and subroutines of the control script. Additionally, the interrupt function of the coding environment became equally difficult to employ, as the keyboard control would not be stopped by the interrupt command. Prolonged control after system interrupts must be accompanied by a manual stop keyboard listener command, and users without prior operating experience might not be able to detect if keyboard control is still activated.

5.2.2 Motor hysteresis

In the context of this thesis, motor hysteresis refers to the influence on linear actuation by the operating direction, duration or control parameters. To understand the differences between tip/tilt motors and whether motor hysteresis has a significant effect, loop scans were performed where a motor would scan along one direction and immediately scan in the opposite direction without setting a new step-zero position.

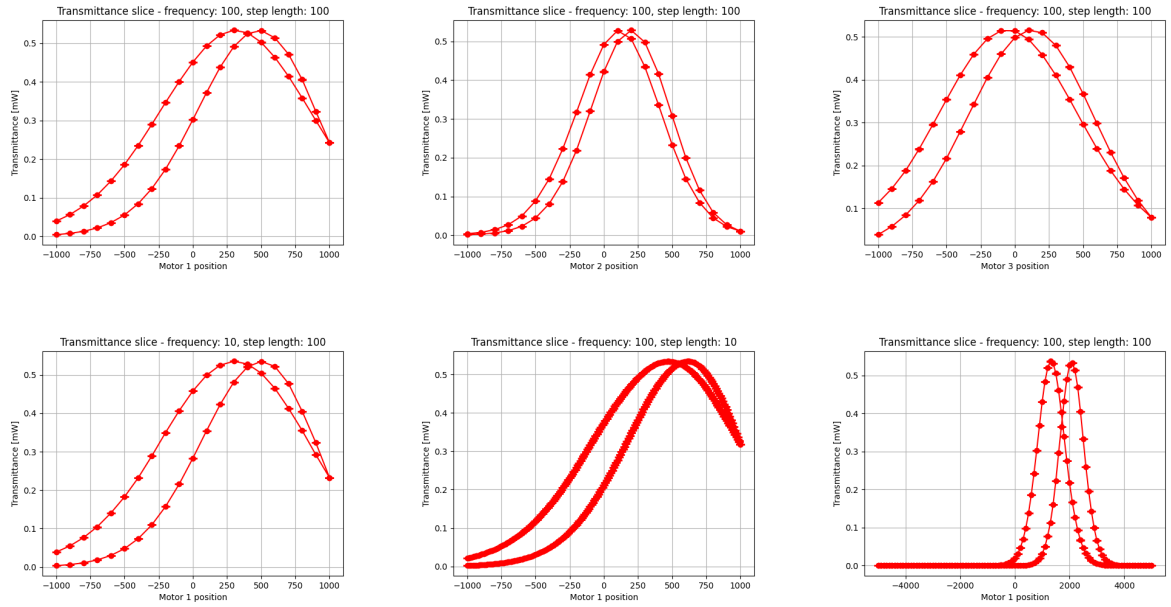


Figure 5.2.2: Example of transmitted power graphs of the rPS with peak power shifts. Top: examples of rPS motors with same parameters, bottom: examples of motor 1 scans for different parameters

The tip motors (left and right graphs in Figure 5.2.2 top row) exhibit a greater transmission window than the tilt motor (centre graph, top row), and the transmission maximum is visibly shifted depending on the scan direction. This is a clear sign of non-negligible motor hysteresis that will affect stability and precision after cryogenic deployment. Different parameters were tested, see Figure 5.2.2 bottom row, to see how they influenced motor hysteresis, as summarized in Table 5.2.1, using motor classifications discussed in Section 5.3.

Motor class	Tip	Tip [%]	Tilt	Tilt [%]
Frequency: 100 Hz Step length: 100 Position range: ±1k	200	10%	100	5%
Frequency: 100 Hz Step length: 10 Position range: ±1k	160	8%	90	4.5%
Frequency: 10 Hz Step length: 100 Position range: ±1k	200	10%	100	5%
Frequency: 10 Hz Step length: 10 Position range: ±1k	190	9.5%	120	6%
Frequency: 100 Hz Step length: 100 Position range: ±5k	2000	20%	800	8%

Table 5.2.1: Maxima peak shift analysis of motor hysteresis.

5.2.3 Internal reflections and non-global maxima

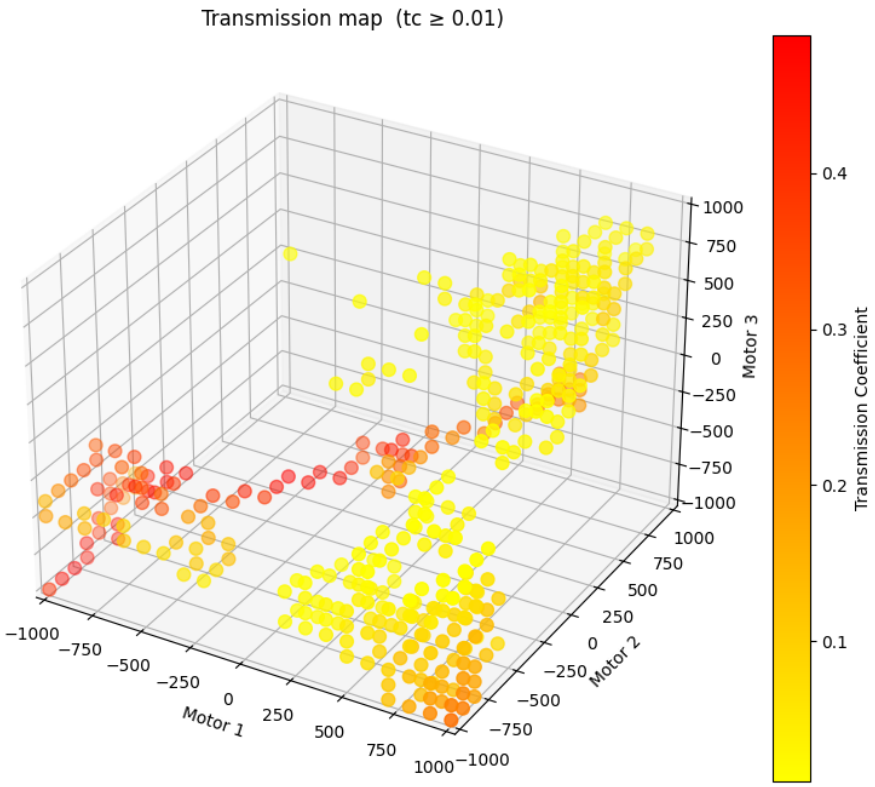


Figure 5.2.3: Transmission coefficient plot for ±1k scan of the rPS. Only transmission coefficients above 1% are included in the plot.

To verify that the system was aligned to the global maximum, tests were conducted to identify the presence of internal reflections. Figure 5.2.3 demonstrates that maximum transmission occurs along the diagonal axis of the motor parameter space. However, the off-diagonal transmission coefficients are non-negligible and exceed the lower-bound estimate of 10% previously proven to result in well-aligned transmission above 50%, determined in section 4.2.1. Furthermore, these long scans showed clear visual evidence of long-term motor hysteresis effects, as the individual motors were positioned farther outward than expected (counter-clockwise movement).

5.3 Results

The highest transmission achieved during this project was 56.75% on 2025-04-14, after an operation time of approximately 8 hr and 50 min. This was achieved by using the procedure outlined in Section 4.2.1 through the COFB-control script in Appendix A, using the 'clever'

gradient cutoff subroutine. The fastest 0% to +50% transmission for single-stage alignment with a well-collimated system was achieved on 2025-04-17 in 9 min and 40 s over 9 alignment iterations.

Motor characterisation

To characterize the motors, the transmitted power profiles are fitted with a Gaussian function, from which the transmission window around the peak is quantified by the FWHM of the fit, see Figure 5.3.1.

Motors 1 and 3 have comparable transmission windows of $\text{FWHM}_1 = 1362$ and $\text{FWHM}_3 = 1272$, while motor 2 has a narrower transmission window of $\text{FWHM}_2 = 786$. The similar values of motor 1-3 FWHMs, and a lower value for motor 2, indicate that a classification scheme of tip or tilt motors could reduce the procedure time and maintain stability, by using different parameters for tip/tilt motors. Data from the ePS indicates similar results; motors 4-6 with similar transmission windows to motors 1-3 and motor 5 with a similar window to 2, but is omitted from this thesis due to lack of complete alignment scans.

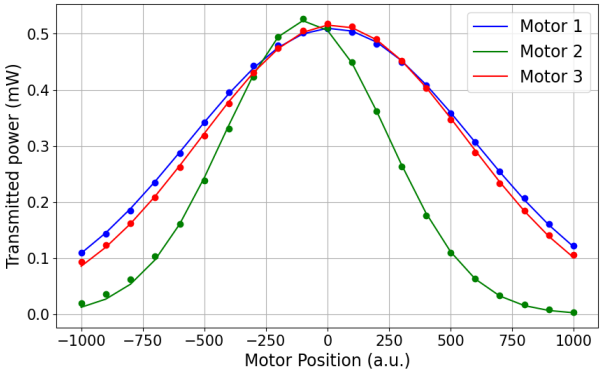


Figure 5.3.1: Gaussian fits for three rPS alignment procedure scans with position range ±1k.

Component characteristics

The power meters failed eight times during the 52 hour scan shown in Figure 5.2.3 (shown as vertical red lines), a considerable improvement from earlier versions of the measurement setup. This improvement is mainly attributed to placing the photo-diodes on magnetic bases. By extension, the primary source of previous errors is believed to have been static electricity from older components on the optical table. The reference diode measured a mean power $\bar{P}_{\text{ref}} = 0.9572\,\text{mW}$ corresponding to a total attenuation, from the incoming power $P_{\text{incoming}} \approx 5\,\text{mW}$ derived from laser source $P_{\text{laser}} \approx 20\,\text{mW}$ with two beamsplitters and

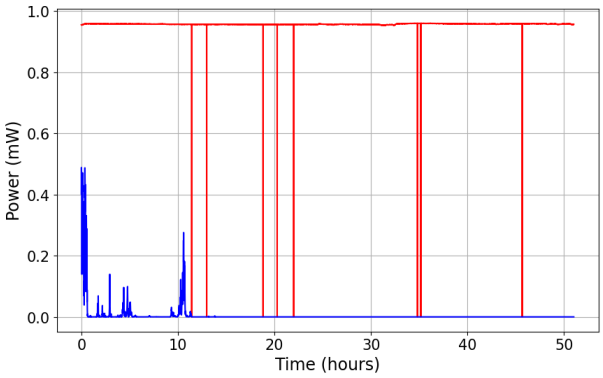


Figure 5.3.2: Reference and transmitted power graph for figure 5.2.3 scan.

negligible loss in mating sleeves, $\alpha = 7.1797\,\mathrm{dB}$. In previous experiments, connecting a multi-mode fibre to the laser sources resulted in over 80% transmission to the reference power meter. Therefore, the increased attenuation loss is most likely due to the single-mode fibre connection to the laser source. The laser output power had STD = 0.0012 mW, with a spread of 0.0067 mW, and a difference between start-up and final power of 0.0017 mW.

Automation of alignment procedure

Following the results produced by the method outlined in Section 4.2.1 a non-interactive script was developed, for aligning the system with well-collimated lenses. A structured hierarchy here referred to as the "order of alignment," is introduced to quantify the precision of the automatic alignment process. There are four orders of alignment defined by the transmission coefficient (Tc), each with its own set of baseline operation parameters following the 10:1:1 standard parameter ratio:

Noise: This order refers to the Tc $\leq$ 15% region. This order corresponds to step (1) in the general alignment procedure and has a baseline position range of $\pm 5\,\mathrm{k}$

Better: This order refers to the 15% $<$ Tc $\leq$ 40% range, and has a baseline position range of $\pm 1\,\mathrm{k}$. Additionally for this order, the 'clever' subroutine is activated.

Fine: This order refers to the 40% $<$ Tc $\leq$ 50% range, and has a baseline position range ± 100. The 'clever' subroutine is still activated for this range.

Hyperfine: This order refers to transmission above 50%, and has a baseline position range of ± 10. For this step, the 'clever' subroutine is deactivated and in the absence of an upper boundary, a limit of 100 iterations is set.

Each set of parameters can be set during operation without having to modify the COFB-control script. When comparing the efficiency of the automated procedure to the manual supervision scheme, the overall alignment duration tends to increase due to additional evaluation steps performed at the beginning and between the automation phases. However, the number of alignment iterations is generally reduced. Both methods reliably produced above 50% transmission when the system was well-aligned when operating the rPS singularly. The fastest automated single-stage alignment from 0% to above 50% transmission was achieved on 2025-05-05 in 12 min and 43 s over 6 alignment iterations.

Chapter 6

Conclusion and Outlook

This thesis discusses the optimization of a Cryogenic optical filter bench (COFB) for the Any Light Particle Search II (ALPS II) experiment, to implement it to reduce the extrinsic Dark-Count Rate (DCR).

Extrinsic dark counts are primarily produced by blackbody radiation from optical couplings of system components situated at room temperature. To reduce blackbody radiation, an ultra-narrow bandpass filter will be deployed at cryogenic temperatures. For signal photons to propagate through the ultra-narrow bandpass filter, the optical output is collimated and subsequently aligned. Due to thermal contraction, a digital control system is implemented to maintain alignment following cryogenic insertion. This is achieved by operating two tip/tilt piston stages, each configured with three linear actuators, commonly referred to as step motors.

The COFB was tested using a 1064 nm tabletop laser source, simulating the experiment parameters of ALPS II, and trial-and-error experiments were conducted to test different methods of controlling the COFB at room temperature. Sequential iterative evaluation of the six parameters proved least likely to align the COFB towards local maxima while ensuring a reasonable scan time and preserving alignment stability. Reducing the motor position range, step length and step frequency, following a 10:1:1 ratio, while dynamically setting a new step-zero reference position between iterations, yielded transmission of above 50%. Reproducibility of high transmission was achieved by repeating a two-step piston stage alignment followed by manual collimation of the rPS until the maximum transmission was reached. Testing the transmission window of each motor yielded two classifications, corresponding to the tip/tilt property of the piston stages. Utilizing this property, further optimization of the procedure could implement motor-specific scanning ranges without extensive software development.

The limiting factor of operation duration and achievable transmission was manually collimating the rPS and ePS, and different methods to optimize this procedure were tested. Visually aligning the collimating lenses inside the collimating cavity provided a simple and effective way to avoid local maxima, but further investigation of low-transmission power graphs with multiple peaks, could enable the development of a system capable of performing the initial collimation within the cryogenic environment.

Tests to evaluate the existence of internal reflection maxima were conducted and provided insights into alignment stability, motor hysteresis and the existence of local maxima above the 10% transmission previously used to confirm alignment towards the global maximum. Based on these findings, additional tests were conducted to increase the understanding of motor hysteresis and its impact on the COFB. Evidence for local transmission maxima was found, but due to extended motor hysteresis during these tests, investigation of these positions showed no local maxima. Whether these local maxima indicate cavity reflections or results from the scanning path should be investigated in future work.

In a well-collimated configuration, transmission above 50% was achieved within a reasonable timespan of less than 10 min, and maintained without losses over an extended period without operation. Under the assumption of minimal losses past the COFB, this performance supports a detection efficiency adequate for attaining the target detection significance of $S = 5$, with the experiment parameters of the ALPS II experiment [19], if the DCR does not exceed 10 µHz and the detection analysis is 90% accurate, as shown in Figure 3.1.3.

6.1 Outlook

Gaussian beam alignment

During system operation, collimating the ePS at long range was generally believed to result in improved beam collimation. Contrary to this assumption, a rigorous analysis of the COFB system based on the thin-lens formula for Gaussian beams Eq. (3.2.10) reveals a different outcome. Considering the system parameters $\lambda = 1064\,\text{nm}$, $f = 8\,\text{mm}$, $w_0 = (6.2/2)\,\mu\text{m}$, and a slight misalignment in the ePS collimating cavity $dx = 1\,\mu\text{m}$, such that the object distance s is $s = f + dx$, then the image distance (to beam waist) $s' = 0.0874\,\text{m}$ and the Rayleigh length $z'_R = 2.2527\,\text{m}$, clearly reveal collimating towards a target much closer could provide a better results. To an approximation, the ePS can be considered well collimated as long as the rPS remains within the Rayleigh length. Considering the geometrics of the Gaussian beam in Figure 3.2.4, the best degree of collimation is likely achieved if the beam waist is centred directly in-between the ePS and rPS. Estimating the distance between the two stages to 5 cm for the beam waist to

be located at the centre of the two stages (image distance $s' = 2.5\,\text{cm}$) the misalignment in the rPS must not exceed $dx = 0.2\,\mu\text{m}$. This would require extreme precision, which is not viable for manual input control.

During the collimation phase, it was generally easier to collimate the lens by approaching the focal point f from large s. This also stands out from Eq. 3.2.10 by considering a slight misalignment on the inside of the focal point, i.e. $dx = -1\,\mu\text{m}$ the image distance is evaluated to be $s' = -63.9126\,\text{m}$ while z'_R remains the same. The virtual image distance, produced when approaching f from small s, is much greater than the image distance, produced when approaching f from large s, which explains the increased sensitivity.

Thermal contraction effects

After cryogenic insertion, the components of the COFB will contract depending on their material composition, as outlined in Section 3.2.1. This will affect the alignment and collimation of the ePS and rPS. Following from the discussion above, if a perfectly collimated lens $s = f = 8\,\text{mm}$ made from titanium with $\alpha_L^{(\text{Ti})} = 8.41 \cdot 10^{-6}\,\text{K}^{-1}$, is cooled down from room temperature $293\,\text{K}$ to cryogenic temperature $40\,\text{K}$ the object distance will be $s_{\text{final}} = 7.9830\,\text{mm}$, corresponding to a $dx = -17\,\mu\text{m}$ shift. Comparing with the results in the above discussion the image distance is shifted to $s' = -2.7623\,\text{m}$ and the Rayleigh length to $z'_R = 1.6598\,\text{m}$, resulting in severe misalignment.

The COFB version operated for the duration of this thesis has a body constructed of Aluminium, but for future versions, the body material is expected to be constructed from a metal alloy known as Invar (64% Fe, 36% Ni) which exhibits a low linear expansion coefficient of $\alpha_L^{(\text{Invar})} = 1.2 \cdot 10^{-6}\,\text{K}^{-1}$ [69], and the piston stages is constructed from titanium. When the ePS is cooled down the beam output is shifted downwards and omitting the collimating lens, is susceptible to misalignment. Approximating the height of the piston stages to be $h_{\text{PS}} = 6.5\,\text{cm}$, then then beam output would be located at $h_{\text{Initial}} = 3.75\,\text{cm}$. Considering the shift in laser output if only a single piston stage was cooled down, the post-cool-down beam output height would be $h_{\text{final}} = 3.7420\,\text{cm}$ corresponding to a shift $dx = -80\,\mu\text{m}$. Comparing this shift with the MFD of the single-mode optical fibre $6.2\,\mu\text{m}$ in the rPS, which for a moment is considered to not thermally contract, the fibres would have no transmission overlap with edge-on distance of $71.8\,\mu\text{m}$. This calculations does not accurately model the physics of system-wide cryogenic cool-down, instead, it serves as a worst-case representation of non-homogeneous thermal effects during the cooling process. The COFB currently lacks translational movement capabilities, and currently, the only way to address these issues is by angling the ePS and rPS. Following this demonstration, the distance between the piston stages in the COFB, if constructed by Invar,

would shorten from 5 cm to $l = 4.9985$ cm. Considering the triangle made from the cryogenic length l between the piston-stages in the COFB, and the cryogenic height shift the angle relation of the ePS and rPS are $\theta = \tan^{-1}\left(\frac{dx}{l}\right) = 0.092°$. Since this angle is within the piston stage tip/tilt angular range of $\pm 5.6°$, aligning the ePS and rPS may be considered as a solution; however, any pre-positioning based solely on these theoretical considerations from the above section is deemed impractical.

Next version of the cryogenic optical fibre bench

Considering thermal effects, the current collimation setup lacks the necessary remote collimation capabilities. These capabilities extend beyond enhancing collimation accuracy and are system-critical for maintaining SDE.

The next version of the cryogenic optical filter bench, shown in Figure 6.1.1, features two translation mounts instead of the coupled lens cavities. This allows for remote control of the object distance s and by extension collimation inside the cryostat. In addition, the piston stages will also be mounted on rail sliders to manually collimate the lenses before cryogenic deployment.

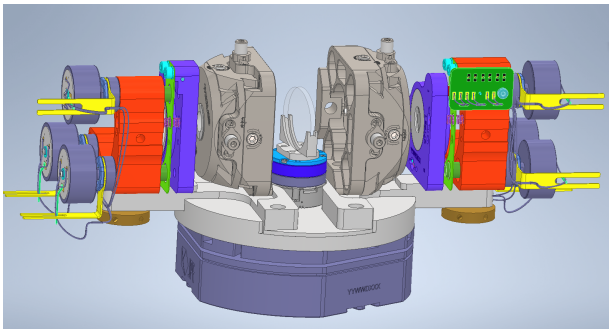


Figure 6.1.1: Future version of the COFB featuring two translational lens mounts and ePS, rPS rail sliders.

6.2 Acknowledgments

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Appendix A

COFB-control script

```
# %% [markdown]
# # Cold Optical Filter Bench - Control script

# %% [markdown]
# ### System setup

# %%
## import libraries to control motors and measurment devices
import sys
import os
import numpy as np
import matplotlib.pyplot as plt
import time
import winsound
import itertools
import random as rn
import subprocess as sp
from IPython.display import display, clear_output

## Import libraries for data handling
import json
from datetime import datetime
from pathlib import Path

## import libraries to use keyboard control
from pynput import keyboard

# Add subdir to the system path
sys.path.append(os.path.abspath('JPE'))
```

```
sys.path.append(os.path.abspath('Thorlabs'))

from JPE_MotorControl import JPEDevice # This is the object corresponding to a certain motor

import Thorlabs_PowerMeasurement as pm # This is the module to access the THORLABS power meters

from TLPM import TLPM # TLPM.py assumed to be in the same directory with this file
os.add_dll_directory(os.getcwd()) # The TLPM_64.dll file is assumed to be in the same directory with this file

# %% [markdown]
# ## Primary functions

# %%
### Saving functions
def get_base_folder(func_name):
    """Create and return base folder path based on date and function name."""
    date_folder = datetime.now().strftime('%Y-%m-%d')
    base_folder = os.path.join('Data', date_folder, f'{func_name}_data')
    os.makedirs(base_folder, exist_ok=True)
    return base_folder

def save_list(my_list, func_name, list_name):
    """Save list as JSON."""
    base_folder = get_base_folder(func_name)
    timestamp = datetime.now().strftime('%H-%M-%S')
    path = os.path.join(base_folder, f'{list_name}_{timestamp}.json')
    with open(path, 'w') as f:
        json.dump(my_list, f)
    print(f"List saved to {path}")

def save_figure(func_name, figure_name, fig=None):
    """Save current or given matplotlib figure."""
    base_folder = get_base_folder(func_name)
    timestamp = datetime.now().strftime('%H-%M-%S')
    path = os.path.join(base_folder, f'{figure_name}_{timestamp}.png')
    if fig is None:
        fig = plt.gcf()
    fig.savefig(path)
    print(f"Figure saved to {path}")

def save_array(np_array, func_name, array_name):
```

```
    """Save NumPy array as .npy file."""
    base_folder = get_base_folder(func_name)
    timestamp = datetime.now().strftime('%H-%M-%S')
    path = os.path.join(base_folder, f'{array_name}_{timestamp}.npy')
    np.save(path, np_array)
    print(f"Array saved to {path}")

## Keyboard control
running = True  # Control flag
pause = False # Pause flag
help = False # Help flag
details = True # Details flag
clever = False # Clever flag
breakout = False # Breakout flag

# Keyboard press key function
def on_press(key):
    """
    Function that takes input from keyboard while script is running and changes the global parameters
    """
    global running
    global pause
    global help
    global details
    global clever
    global breakout
    try:
        if key.char == 's':  # If 's' is pressed
            print("Stopping script...")
            running = False

        if key.char == 'b':  # If 's' is pressed
            print("Breaking out of loop...")
            breakout = True

        if key.char == 'p': # If 'p' is pressed
            if pause == True:
                print('Unpausing...')
                pause = False
            elif pause == False:
                print('Pausing...')
                pause = True

```

```
114         if key.char == 'h': # if 'h is pressed
115             if help == False:
116                 print('Acquiring help...')
117                 help = True
118             elif help == True:
119                 print('Dismissing help...')
120                 help = False
121 
122         if key.char == 'd': # if 'd' is pressed
123             if details == True:
124                 print('Turning off details...')
125                 details = False
126             elif details == False:
127                 print('Turning on details...')
128                 details = True
129 
130         if key.char == 'c': # if 'c' is pressed
131             print('"c" has been pressed...')
132             if clever == True:
133                 clever = False
134             elif clever == False:
135                 clever = True
136 
137     except AttributeError:
138         pass  # Ignore special keys
139 
140 
141 ## Power meter measuring functions
142 measuring_time = 0.75 # Measurement time (s)
143 
144 def measure_power(device_name_str='USB0::0x1313::0x807B::230824323::INSTR',
    measurement_time=measuring_time, time_step=0.05, operating_wave_legth=1065):
145     '''
146     Confirm that the deice_name_str corresponds to the correct measurement!
147 
148     Measures the power transmitted throught the filter bench
149     OUTPUT: average_power, std
150     '''
151     ts, powers = pm.measure_power(device_name_str, measurement_time, time_step,
    operating_wave_legth)
152     return np.average(powers), np.std(powers)
153 
154 
```

```
155 def measure_reference_power(device_name_str='USB0::0x1313::0x807B::230920224::INSTR'
    , measurement_time=measuring_time, time_step=0.05, operating_wave_legth=1065):
156     '''
157     Confirm that the deice_name_str corresponds to the correct measurement!
158 
159     Measures the power transmitted throught the filter bench
160     OUTPUT: average_power, std
161     '''
162     ts, powers = pm.measure_power(device_name_str, measurement_time, time_step,
    operating_wave_legth)
163     return np.average(powers), np.std(powers)
164 
165 
166 ## Transmission Coefficient function
167 def transmission_coefficient():
168     """
169     Uses the measure_reference_power and measure_power functions to determine the
    transmission coefficient.
170     OUTPUT: t (transmission coefficient) [%]
171     """
172     try:
173         p_trans, err = measure_power()
174         p_ref, err = measure_reference_power()
175 
176         t = round(100*p_trans/p_ref, 3)
177     except:
178         print('Error occured...')
179         t = 0
180 
181     return t
182 
183 ## Motor control
184 # First piston stage / recieving piston stag e(CTTPS #1):
185 CTTPS1_axis1 = JPEDevice(COMport_nb=4, address=1, stage_type='CTTPS1')
186 CTTPS1_axis2 = JPEDevice(COMport_nb=4, address=2, stage_type='CTTPS1')
187 CTTPS1_axis3 = JPEDevice(COMport_nb=4, address=3, stage_type='CTTPS1')
188 
189 # Second piston stage / emitting piston stage (CTTPS #2):
190 CTTPS2_axis1 = JPEDevice(COMport_nb=3, address=1, stage_type='CTTPS1')
191 CTTPS2_axis2 = JPEDevice(COMport_nb=3, address=2, stage_type='CTTPS1')
192 CTTPS2_axis3 = JPEDevice(COMport_nb=3, address=3, stage_type='CTTPS1')
193 
194 # Rotator (CSR1):
195 #CSR = JPEDevice(COMport_nb=4, address=4, stage_type='CSR1')
```

```
196
197
198 # Motor list (CTTPS#1 + CTTPS#2)
199 all_motors = [CTTPS1_axis1, CTTPS1_axis2, CTTPS1_axis3, CTTPS2_axis1, CTTPS2_axis2,
        CTTPS2_axis3]
200
201 # Piston stage 1 (reciving stage) and 2 (emitting stage)
202 stage_1 = [all_motors[0], all_motors[1], all_motors[2]]
203 stage_2 = [all_motors[3], all_motors[4], all_motors[5]]
204
205
206
207 ## Motor identification
208 def identify_motor(motor):
209     """
210     Function to identify motor call nr from motor string.
211     OUTPUT: motor #
212     """
213     if motor is CTTPS1_axis1:
214         return 1
215     elif motor is CTTPS1_axis2:
216         return 2
217     elif motor is CTTPS1_axis3:
218         return 3
219     elif motor is CTTPS2_axis1:
220         return 4
221     elif motor is CTTPS2_axis2:
222         return 5
223     elif motor is CTTPS2_axis3:
224         return 6
225     else:
226         return "Unknown motor"
227
228
229
230
231 # %% [markdown]
232 # ## Secondary functions
233
234 # %%
235 ## Manual zero-step value positioning
236 def manual_zero_pos(freq = 600):
237     """
```

```
    Manually adjust the position of motors. During run the system will require
    inputs to define the motor which is in operation

    Functions used:
    identify_motor()

    INPUT:
    freq (int): Movement speed of the motor [Hz] set to 600Hz unless otherwise
    defined

    OUTPUT:
    """
    # Define global variables used by this script
    global running  # Used to stop operation of code
    global clever   # Used to change motor during operation

    # Start keyboard 'listener' in a separate thread
    listener = keyboard.Listener(on_press = on_press)
    listener.start()
    try:
        print('Choose motor: ')
        motor = all_motors[int(input('Choose motor:')) - 1]
        clear_output()
        while running == True:

            print(f'Adjusting motor {identify_motor(motor)}')
            print('Input new position: (inwards/clockwise = +, outwards/
    counterclockwise = -)')
            new_position = int(input("New position"))

            # Sets position to input
            motor.set_position(new_position, step_size=100, freq=freq, temp=293,
    drive_factor=1.0, print_device_output=False, safemode=False)
            clear_output()

            # Sets current position to step-value zero position
            motor.set_new_zero_position()

            # Ask for input to change motor or stop. If no input is detected
    operation continues for same motor
            print('Press "c" to change motor')
            print('Press "s" to stop')
            time.sleep(5)
            clear_output()
```

```
277
278             if clever == True:
279                 print('Choose new motor:')
280                 motor = all_motors[int(input('Choose motor:')) - 1]
281                 clever = False
282                 clear_output()
283     except:
284         print('Error occured')
285
286     # After finishing code set running to true for continuous system interaction
287     running = True
288
289     # Stop keyboard control
290     listener.stop()
291     print('Script finished.')
292
293 ## Automatic iterative hotspot function
294 def Auto_iterative_hotspot(motors, min_pos, max_pos, step_length, iterations,
    shuffle = False, threshold = None, freq = 600):
295     """
296     Automatic iterative scanning function with maxima step-zero positioning
297
298     Functions used:
299     saving functions
300     identify_motor()
301
302     INPUTS:
303     motors (list/variable): Motors or motor if singular input, to operate
304     min_pos (int): Minimum position (step-value)
305     max_pos (int): Maximum position (step-valu)
306     step_length (int): Distance between position in a list from min_pos to max_pos
307     iterations (int): Nr. of iterations
308     shuffle (bool) = False: shuffles the motors position if defined
309     threshold (double) = transmission coefficient threshold [%]
310
311     OUTPUTS:
312     """
313     # Start keyboard 'listener' in a separate thread
314     listener = keyboard.Listener(on_press=on_press)
315     listener.start()
316
317     # Start timer
318     start_time = time.time()
319
```

```
320     # Define global variables used by this script
321     global running  # Used to stop operation
322     global pause    # Used to pause operation
323     global help     # Used to show details of possible keyboard inputs
324     global details  # Used to toggle details of current scan
325     global clever   # Used to toggle early cutoff with negative gradient
326 
327     # Error analysis
328     err_scan = 0
329     err_set_optimal_pos = 0
330     err_transmission_coefficient = 0
331 
332 
333     # If single motor make motors variable into a list
334     if not isinstance(motors,list):
335         motors = [motors]
336 
337     # Define position list for motor input
338     positions = list(range(min_pos, max_pos + 1, step_length))
339 
340     # Predefine data lists
341     Ts = []     # Transmission coefficients
342     times = []  # Times for each motor iteration
343 
344     # Run scan for each iteration in range of total iterations
345     for iter in range(iterations):
346 
347         # Check stop signal
348         if running == False:
349             running = True
350             break
351 
352         # Randomize the order of optimization:
353         if shuffle is True:
354             rn.shuffle(motors)
355 
356         # show iteration nr.
357         print(f'Iteration: {iter + 1}')
358 
359         for motor in motors:
360             # start motor time measurement
361             motor_start_time = time.time()
362 
363             # Identify motor
```

```
            motor_id = identify_motor(motor)

            # Check for stop signal
            if running == False:
                break

            # Define empty list for current scan
            measured_powers = []
            errs = []
            xs = []

            # Run scan for position values in positions list
            for pos in positions:

                # Set motor to current position in list
                motor.set_position(pos, step_size=100, freq=freq, temp=293, drive_factor=1.0, print_device_output=False, safemode=False)

                # Try to measure power (disconnection issues may provoke error)
                try:
                    p, err = measure_power()

                    # Store transmitted power and std in predefined lists
                    measured_powers.append(p)
                    errs.append(err)

                # If issue with power meters
                except:
                    # Error analysis data
                    err_scan = err_scan + 1

                    # Store a zero-value in place of transmitted power
                    measured_powers.append(0)
                    errs.append(0)
                    print('ERROR OCCURED !!!')

                # Store the coresponding list
                xs.append(pos)

                # Plot the tranmistted power and position. Updates during loop for each measurenmet (only works in a .ipynb environment).
                plt.errorbar(xs, measured_powers, yerr=errs, fmt='o', ecolor='r', capsize=5, capthick=2, linestyle='-', color='red')
                plt.xlabel(f'Motor {motor_id} position (a.u.)')
```

```
405                 plt.ylabel('Transmittance (mW)')
406                 clear_output()
407                 plt.show()
408 
409                 # Check for pause signal
410                 if pause == True:
411                     print('Paused.')
412                     while pause == True:
413                         time.sleep(1)
414 
415 
416                 # Check for toggle help signal
417                 if help == True:
418                     # Show detail of usable keyboard control for this script
419                     print('Press "h" to toggle help.')
420                     print('Press "s" to stop after this motor.')
421                     print('Press "p" to pause.')
422                     print('Press "d" to toggle details.')
423                     print('Press "c" to toggle clever mode.')
424 
425                 # Check for toggle details signal
426                 if details == True:
427                     print(f'Iteration: {iter + 1}')
428                     if clever == True:
429                         print('Clever mode is on.')
430                     if Ts:
431                         print(f'Last evaluated transmission coefficient {t}%')
432                     print(f'Errors: scan = {err_scan}, optimal position
    configuration = {err_set_optimal_pos}, transmission coefficient = {
    err_transmission_coefficient}')
433 
434                 # Check for stopping signal
435                 if running == False:
436                     print('Stopping...')
437 
438                 # If clever is on stop script early
439                 if  clever == True and len(measured_powers) >= 2:
440 
441                     # Secondary parameters to validate a negative gradiant is not a
    std mistake
442                     if measured_powers[-1] != 0 and measured_powers[-1] <=
    measured_powers[-2] and abs(measured_powers[-1]-measured_powers[-2]) > errs[-1]
    and abs(measured_powers[-1]-measured_powers[-2]) > errs[-2]:
```

```
443                     print('Cleverly stopping position evalutation for this code.
    ')
444                     break
445
446
447
448
449         # Stop timer for motor iterations and save to a list
450         motor_stop_time = time.time()
451         motor_run_time = motor_stop_time - motor_start_time
452         times.append(motor_run_time)
453
454         # Find maximum tranmistted power and corresponding position
455         optimal_position = positions[measured_powers.index(max(measured_powers))
     ]
456         print('Optimal position:', optimal_position)
457
458         # Print the scan and plot a line at maximum transmission
459         plt.axvline(x=optimal_position, color='black', linestyle='--')
460         plt.errorbar(xs, measured_powers, yerr=errs, fmt='o', ecolor='r',
    capsize=5, capthick=2, linestyle='-', color='red')
461         plt.xlabel(f'Motor {motor_id} position (a.u.)')
462         plt.ylabel('Transmittance (mW)')
463         clear_output()
464
465         # Save figure and list
466         save_figure('AIH', 'transmittance')
467         save_list(measured_powers, 'AIH', 'transmittance')
468         plt.show()
469
470         # Set motor to optimum position
471         print('Setting motor to optimal position')
472         try:
473             motor.set_position(optimal_position, step_size=100, freq=freq, temp
    =293, drive_factor=1.0, print_device_output=False, safemode=False)
474             motor.set_new_zero_position()
475         except:
476             err_set_optimal_pos = err_set_optimal_pos + 1
477             print('ERROR OCCURED !!!')
478
479         # Calculate transmission coefficient
480         try:
481             # Measure transmission coefficient (this could also be done using
    transmission_coefficient() function)
```

```
                p_trans, err = measure_power()
                p_ref, err = measure_reference_power()
                t = round(100*p_trans/p_ref, 2)

                # Store transmission coefficient in list
                Ts.append(t)

            except:
                err_transmission_coefficient = err_transmission_coefficient + 1
                print('Error occured')
                Ts.append(0)


            # plot transmission coefficient and motor iterations
            plt.plot(Ts, 'o-')
            plt.ylabel('Transmission Coeff.')
            plt.xlabel(f'Motor iteration (1/{len(motors)} full iteration)')
            plt.grid(True)
            clear_output()
            plt.show()
            time.sleep(3)

            # Check for pause signal
            if pause == True:
                print('Paused.')
                while pause == True:
                    time.sleep(1)

            # If Threshold is set stop if transmission coefficient is at or above threshold.
            if threshold is not None and t >= threshold:
                print('Threshold reached, stopping script...')
                running = False
                break

    # Stop listening for keyboard inputs
    listener.stop()

    # Calculate total run time
    stop_time = time.time()
    run_time = stop_time - start_time
    times.append(run_time)

    # save times
```

```
525     save_list(times, 'AIH', 'times')
526     clear_output()
527 
528     # Plot final transmission coefficient list and motor iterations to save
529     plt.plot(Ts, 'o-')
530     plt.ylabel('Transmission Coeff.')
531     plt.xlabel(f'Motor iteration (1/{len(motors)} full iteration)')
532     plt.grid(True)
533     save_figure('AIH', 'transmission')
534     save_list(Ts, 'AIH', 'transmission')
535     plt.show()
536 
537     # Print details from alignment script
538     print(f'Total iterations: {iter} - Total run time: {run_time:.1f}')
539     print("Script finished.")
540 
541     # Make a sound
542     winsound.Beep(1000, 1500)
543 
544 
545 
546 ## Transmittance slice function
547 def transmittance_slice(motor, min_pos, max_pos, step_length, normal = True, reverse
        = False, turnaround = False, freq = 600):
548     """
549     Function to get transmitted power curve for single motor, and manually input new
        step-zero position afterwards.
550 
551     Functions used:
552     saving functions
553     identify_motor()
554 
555     INPUTS:
556     motor (variable): the motor to use in creating the transmitted power curve
557     min_pos (int): Minimum position (steps)
558     max_pos (int): Maximum position (steps)
559     step_length (int): Distance betweem position in a list from min_pos to max_pos
560     normal (bool): True - function is run as normal scan, False - function is run as
        an analysis
561     reverse (bool): True - reverses position list so can is run for max -> min
562     turnaround (bool): True - scan runs from min -> max -> min
563     freq (int): Movement speed [Hz]
564 
565     OUTPUTS
```

```
    """
    # Start keyboard 'listener' in a separate thread
    listener = keyboard.Listener(on_press=on_press)
    listener.start()

    # Define global variables used by this script
    global running  # Used to stop operation
    global pause    # Used to pause operation
    global help     # Used to show details of possible keyboard inputs

    # Identify motor
    motor_id = identify_motor(motor)

    # Construct positions
    positions = list(range(min_pos, max_pos + 1, step_length))

    # Run script for reversed position list (from max -> min)
    if reverse is True:
        positions.reverse()
        print('Positions reversed')

    # Define scanning parameter lists
    measured_powers = []
    errs = []
    pos = []

    # Check for number of fails:
    fails = 0

    # Activate running parameter:
    slicing = True
    while slicing is True:

        # Run scan for each position value in position list
        for position in positions:
            try:
                ## Set motor position at given coordinate and measure the power and
    std
                motor.set_position(position, step_size=100, freq=freq, temp=293,
    drive_factor=1.0, print_device_output=False, safemode=True)
                power, err = measure_power()

            except:
                # If failed note for error analysis
```

```
608                 print('Error occured!')
609                 fails = fails + 1
610 
611             # Put measured power, std and position into lists
612             measured_powers.append(power)
613             errs.append(err)
614             pos.append(position)
615 
616             # Create plot showing the transmitted power curve. Updates during loop
    for each measurenmet (only works in a .ipynb environment).
617             plt.errorbar(pos, measured_powers, yerr = errs, fmt='o', ecolor='r',
    capsize=5, capthick=2, linestyle='-', color='red')
618             plt.xlabel(f'Motor {motor_id} position')
619             plt.ylabel('Transmittance [mW]')
620             plt.grid(True)
621 
622             clear_output()
623             plt.show()
624 
625             # Check for pause signal
626             while pause == True:
627                 print('Paused.')
628                 # Plot scan while paused
629                 plt.errorbar(pos, measured_powers, yerr = errs, fmt='o', ecolor='r',
     capsize=5, capthick=2, linestyle='-', color='red')
630                 plt.xlabel(f'Motor {motor_id} position')
631                 plt.ylabel('Transmittance [mW]')
632                 plt.grid(True)
633 
634                 plt.show()
635 
636                 # Check for help signal
637                 if help == True:
638                     print('Press "h" to toggle help')
639                     print('Press "s" to stop')
640                     print('Press "p" to unpause')
641                 time.sleep(3)
642                 clear_output()
643 
644             # Check for stopping signal
645             if running == False:
646                 running = True
647                 break
648 
```

```
649         # Check for parameters for hysteresis anaylsis
650         if turnaround is True:
651             positions.reverse()
652             normal = False
653             turnaround = False
654         else:
655             # If no analysis stop function
656             slicing = False
657 
658     # Show amount of fails
659     print(f'Fails = {fails}')
660 
661     # Find maxima if no analysis is done (creates error as one position wil have
    multible measured powers)
662     if normal is True:
663         p_max = max(measured_powers)
664         p_max_index = measured_powers.index(p_max)
665         pos_max = pos[p_max_index]
666 
667 
668     # Creat plot for tranmitted power curve
669     plt.errorbar(pos, measured_powers, yerr = errs, fmt='o', ecolor='r', capsize=5,
    capthick=2, linestyle='-', color='red')
670     plt.xlabel(f'Motor {motor_id} position')
671     plt.ylabel('Transmittance [mW]')
672     plt.title(f'Transmittance slice - frequency: {freq}, step length: {step_length}'
    )
673     plt.grid(True)
674     clear_output()
675 
676     # Save positions and measured powers list.
677     save_list(pos,'slice',f'motor_{motor_id}_pos')
678     save_list(measured_powers,'slice',f'motor_{motor_id}_measured_powers')
679 
680     # Save figure of tranmistted power
681     save_figure('slice',f'motor_{motor_id}_fig')
682 
683     # Show figure
684     plt.show()
685 
686     # Stop keyboard 'listener'
687     listener.stop()
688 
689     # If no analysis is done ask for position to set as new step-zero position
```

```
690     if normal is True:
691         while True:
692             # Show max tranmitted power position
693             print(f'Max transmittance at {pos_max}')
694 
695             # Ask to set new step-zero position
696             print('Set new motor zero position?')
697             ask = input('Set new motor zero position? (yes/no)')
698             if ask == "yes":
699                 # Input new step-zero position
700                 print('Input new position')
701                 new_pos = int(input("New position?"))
702 
703                 # Move motor to new postiion and set as step-zero position
704                 motor.set_position(new_pos, freq=freq, safemode = True)
705                 motor.set_new_zero_position()
706                 break
707 
708             elif ask == "no":
709                 # Reset motor to zero postion
710                 motor.set_position(0, freq=freq, safemode=True)
711                 print('Motor set to initial position')
712                 break
713 
714             else:
715                 # If keyboard input string not defined in control loop try again
716                 print('Not allowed input, please try again.')
717     else:
718         # Resetting motor to zero position
719         motor.set_position(0, freq=freq, safemode=True)
720         print('Motor set to initial position')
721 
722     # Finish script
723     print('Script finished')
724 
725     # Make sound
726     winsound.Beep(1000, 1500)
727 
728 ## Automatic alignment
729 def auto_stage_alignment(stage, pos_1, step_1, freq_1, pos_2, step_2, freq_2, pos_3,
        step_3, freq_3, pos_4, step_4, freq_4, hyperfine_iterations):
730     """
731     Automatic aligment function for a well-collimated cold optical filter bench
732     Nr. in parameters refer to order of alignment
```

```
    1 - noise alignment
    2 - better alignment
    3 - fine alignment
    4 - hyperfine alignment

    Functions used:
    saving functions
    transmission_coefficient()
    Auto_iterative_hotspot()

    INPUTS:
    stage (any): Which stage to align, or all_motors
    pos_# (int): Position range (convention minimum position = - pos, maximum position = pos) for order #
    step_# (int): Distance between position in a list from min_pos to max_pos for order #
    freq_# (int): Movement speed [Hz] for order #
    hyperfine_iterations (int): Nr of iterations the system will do at hyperfine order before stopping
    """

    # Define global variables used by this script
    global running  # Used to stop operation for a single automation step
    global breakout # Secondary operation stop for automatic script
    global clever   # Used to toggle early cutoff with negative gradient

    ## System setup
    # Rough alignment
    min_pos_1 = -pos_1
    max_pos_1 = pos_1
    step_length_1 = step_1
    freq_1 = freq_1

    # Better alignment
    min_pos_2 = -pos_2
    max_pos_2 = pos_2
    step_length_2 = step_2
    freq_2 = freq_2

    # Fine alignment
    min_pos_3 = -pos_3
    max_pos_3 = pos_3
    step_length_3 = step_3
    freq_3 = freq_3
```

```
774
775     # Hyperfine alignment
776     min_pos_4 = -pos_4
777     max_pos_4 = pos_4
778     step_length_4 = step_4
779     freq_4 = freq_4
780
781     # Start timer
782     time_start = time.time()
783
784     # Transmission coefficient list
785     tcs = []
786
787     # Rough alignment (0.01%)
788     t_c = transmission_coefficient() # Measured initial transmission coefficient
789     tcs.append(t_c) # Stores initial transmission coefficient
790
791     # Checks if automation is stopped
792     if breakout == False:
793
794         # Checks if initial transmission is in the noise order
795         while t_c <= 15:
796
797             # Turns of clever for noise order
798             clever = False
799
800             # Runs alignment scheme at order parameters for chosen motors
801             Auto_iterative_hotspot(stage, min_pos_1, max_pos_1, step_length_1,
    iterations = 1, threshold =  15, freq = freq_1)
802
803             # Notes order
804             order = 'Noise'
805             print(f'Aligning {order} order.')
806
807             # Measures and stores transmission coefficient after alignment
808             t_c = transmission_coefficient()
809             tcs.append(t_c)
810
811             # If automation breakout flag is true stops automation step
812             if breakout == True:
813                 print('Stopping code...')
814                 break
815
816
```

```
    # Better alignment (15.0% -> 50.0%)
    # Checks if automation is stopped
    if breakout == False:

        # Checks if last evaluated transmission coefficient is better order
        while t_c <= 40.00:

            # Sets clever equal to true for better alignment
            clever = True

            # Runs alignment scheme at order parameters for chose motors
            Auto_iterative_hotspot(stage, min_pos_2, max_pos_2, step_length_2,
    iterations = 1, threshold =  40, freq = freq_2)

            # Notes order
            order = 'Better'
            print(f'Aligning {order} order.')

            # Measures and stores transmission coefficient after aligment
            t_c = transmission_coefficient()
            tcs.append(t_c)

            # If automation breakout flag is true stops automation step
            if breakout == True:
                print('Stopping code...')
                break


    # fine alignment (+40.0%)
    # Checks if automation is stopped
    if breakout == False:

         # Checks if last evaluated transmission coefficient is fine order
        while t_c <= 50.00:

            # Sets clever equal to true for fine alignment
            clever = True

            # Runs alignment scheme at order parameters for chose motors
            Auto_iterative_hotspot(stage, min_pos_3, max_pos_3, step_length_3,
    iterations = 1, threshold = 50, freq = freq_3)

            # Notes order
            order = 'Fine'
```

```
            print(f'Aligning {order} order.')

            # Measures and stores transmission coefficient after aligment
            t_c = transmission_coefficient()
            tcs.append(t_c)

            # If automation breakout flag is true stops automation step
            if breakout == True:
                print('Stopping code...')
                break


    # Stops measuring time to evaluate automation alignment efficiency
    time_stop = time.time()

    # hyperfine alignment (+50.0%)
    # Checks if automation is stopped
    if breakout == False:

        # Checks if last evaluated transmission coefficient is hyperfine order
        if t_c >= 50.00:

            # Sets clever equal to true for hyperfine alignment
            clever = False

            # Runs alignment scheme at order parameters for chose motors
            Auto_iterative_hotspot(stage, min_pos_4, max_pos_4, step_length_4,
    iterations=hyperfine_iterations, freq = freq_4)

            # Notes order
            order = 'Hyperfine'
            print(f'Aligning {order} order.')

            # Measures and stores transmission coefficient after aligment
            t_c = transmission_coefficient()
            tcs.append(t_c)

    # Calculate the total run time in seconds for automated alignment
    run_time = time_stop - time_start

    # Measures and stores final transmission coefficient
    t_f = transmission_coefficient()
    tcs.append(t_f)

```

```
902     # Save list of tranmission coefficients
903     save_list(tcs,'ASA','transmission_coefficients')
904 
905     clear_output()
906 
907     # Print details of automated alignment
908     print(f'Aligment of order: {order}')
909     print(f'Time to reach hyperfine alignment: {run_time}s / {round(run_time/60)}min
        ')
910     print(f'Final transmission coefficient: {t_f}')
911 
912     # Plot the automation transmission coefficient between each alignment step
913     plt.plot(tcs, 'o-')
914     plt.ylabel('Transmission Coeff. for ASA')
915     plt.xlabel(f'Motor iterations')
916     plt.grid(True)
917 
918     # Save figure
919     save_figure('ASA', 'transmission_coefficients')
920 
921     # Show plot
922     plt.show()
923 
924     # Set breakout to False for continued system control
925     breakout = False
926 
927 
928 # %% [markdown]
929 # ## Control Script
930 
931 # %%
932 ## Control scropt for Cold Optical Filter Bench
933 def COFB_control():
934     """
935     Script to control the cold optical filter bench through keyboard input.
936     """
937     print('Initializings COFB conttrol')
938 
939     # Define global variables
940     global running  # Used to stop operation
941     global pause    # Used to pause operation
942     global help     # Used to show details of possible keyboard inputs
943 
944     while running == True:
```

```
945         # Control options
946         print('Which function would you like to use?')
947         print('Manual zero position: manualzero')
948         print('Slicing function: slice')
949         print('Transmission coefficient: trans')
950         print('Automatic Iterative Hotspot: AIH')
951         print('Automatic stage alignment: ASA')
952         print('Exit program: ext')
953 
954         # Input string
955         ch = input('Choose function:')
956 
957         # String to function operator
958         if ch == "manualzero":
959             # Settings for manual_zero_pos() function
960             clear_output()
961 
962             print('Choose frequency')
963             while True:
964                 # Input frequncy
965                 freq = int(input('Frequency:'))
966                 if 1 <= freq <= 600: # Check if frequency is within allowed values
967                     break
968                 else:
969                     print('Frequency not within allowed range, please choos another
    frequency')
970 
971             # Run manual zero positioning function at choosen frequency
972             manual_zero_pos(freq = freq)
973 
974         elif ch == "slice":
975             # Settings for single transmitted power graph and position control (
    referred to as slice)
976 
977             clear_output()
978             # Choose motor nr to use
979             print('Choose motor to slice')
980             while True:
981                 i = int(input('Choose motor (nr):'))
982                 if 1 <= i <= 6: # If motor nr is an allowed motor
983                     # Pick out motor from all_motors list
984                     motor = all_motors[i-1]
985 
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986                     # Test to make sure correct motor is chosen (if shuffle has been
    used earlier this might have changed)
987                     motor_id = identify_motor(motor)
988                     print(f'Motor {motor_id} choosen.')
989                     break
990                 else:
991                     print('Choosen motor does not exits, please choose a motor from
    1 to 6')
992 
993             # Choose position, step length and frequency to use in slicer
994             while True:
995                 print('Choose maxi and mini position:')
996 
997                 # Choose maximum position step-value
998                 max_pos = int(input('Input max position:'))
999 
1000                # Choose minimum position step-value (remember by convention this is
    accompanied by a negative sign)
1001                min_pos = int(input('Input min positions (remember -):'))
1002 
1003                # Define total step range
1004                maximum_step_range = (abs(min_pos)+abs(max_pos))/2
1005 
1006                print('Choose step length:')
1007                step_length = int(input('Input step_length'))
1008 
1009                print('Choose frequency:')
1010                freq = int(input('Choose freq:'))
1011 
1012                # Test that choosen parameters are allowed by the motor-control and
    no discrepancies in choices might cause error in script
1013                if 1 <= freq <= 600 and 1 <= step_length <= maximum_step_range and 0
    <= max_pos <=40000 and -40000 <= min_pos <=0:
1014                    print(f'Choosen control parameters for slicing function: max/min
    = {max_pos}, step length = {step_length}, frequency = {freq}')
1015                    break
1016                else:
1017                    print('Some parameters are illegal, please input slicing
    parameters again.')
1018 
1019            # Choose if slice should be reversed or run in a loop for analysis
    purposes
1020            reverse = False     # If set to True runs script for max -> min
1021            turnaround = False  # If set to True rune script for min -> max -> min
```

```
            while True:
                print('Should the be run normal, reverse or in loop?')
                check = input('normal/reverse/loop')
                if check == "reverse":
                    reverse = True
                    print('Slicing order reversed...')
                    break
                elif check == "loop":
                    turnaround = True
                    print('Slicing one loop...')
                    break
                elif check == "normal":
                    print('Slicing running normal...')
                    break
                else:
                    print('Illegal input, please try again...')

            # Run singular motor transmitted power graphing function
            transmittance_slice(motor ,min_pos, max_pos, step_length, reverse=
    reverse, turnaround=turnaround, freq=freq)

        elif ch == "trans":
            # Evaluate the transmission coefficient
            clear_output()
            try:
                T = transmission_coefficient()
                print(f'Transmission coefficient: {T}%')
            except:
                print('Error occured')
                pass

        elif ch == "AIH":
            # Setting for auto_iterative_hotspot() function
            clear_output()

            # Define motors to run code
            print('Choose motors:')
            while True:
                im = input('Choose motors: (stage_1/stage_2/all_motors)')
                if im == "stage_1":
                    motors = stage_1
                    print('Running script on stage 1.')
                    break
                elif im == "stage_2":
```

```
                    motors = stage_2
                    print('Running script on stage 2..')
                    break
                elif im == "all_motors":
                    motors = all_motors
                    print('Running script on all motors')
                    break
                else:
                    print('Input error! Please choose between stage_1, stage_2 or all_motors')

            ## Define parameters to determine positions
            print('Now choose script parameters.')
            while True:
                print('Choose max position (assuming min pos = - max pos):')
                maxpos = int(input('Choose max position: '))

                # Sets minimum position equal to the negative maximum position for a symmetric scan
                minpos = - maxpos

                print('Choose step length:')
                step_length = int(input('Choose step length'))

                print('Choose frequency:')
                freq = int(input('Choose frequency'))

                print('Choose nr of iterations:')
                iterations = int(input('Nr of iteratios:'))

                 # Test that choosen parameters are allowed by the motor-control and no discrepancies in choices might cause error in script
                if 1 <= freq <= 600 and 1 <= step_length <= maxpos and 1 <= maxpos <= 40000 and iterations >= 1:
                    print(f'Choosen script parameters for AIH: max/min = {maxpos}, step length = {step_length}, frequency = {freq}, iterations = {iterations}')
                    break
                else:
                    print('Some parameters are illegal, please redo script parameters.')

            print(f'Starting AIH script')
            # Run the auto_iterative_hotspot with choosen parameters
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```
            Auto_iterative_hotspot(motors, minpos, maxpos, step_length, iterations,
    freq = freq)

        elif ch == "ASA":
            # Settings for automatig stage alignment function
            clear_output()
            print('Choose stage or all motors to align')
            while True:
                im = input('Choose stage: (stage_1/stage_2/all_motors)')
                if im == "stage_1":
                    stage = stage_1
                    print('Running script on stage 1.')
                    break
                elif im == "stage_2":
                    stage = stage_2
                    print('Running script on stage 2..')
                    break
                elif im == "all_motors":
                    stage = all_motors
                    print('Running script on all_motors')
                    break
                else:
                    print('Input error! Please choose between stage_1, stage_2')

            time.sleep(1.5)
            clear_output()
            ## Baseline values for ASA function
            pos_1 = 5000
            step_1 = 500
            freq_1 = 500

            pos_2 = 1000
            step_2 = 100
            freq_2 = 100

            pos_3 = 100
            step_3 = 10
            freq_3 = 10

            pos_4 = 10
            step_4 = 1
            freq_4 = 1
            hyperfine_iterations = 100

```

```
1145            while True:
1146                print('Use base alignment procedure?')
1147                base = input('yes/no')
1148
1149                if base == 'yes':
1150                    print('Running Automatic single stage alignment.')
1151                    break
1152
1153                elif base == 'no':
1154                    # Adjusting the four individual steps of the automatic process
1155                    print('Adjusting alignment procedure...')
1156
1157                    while True:
1158                        print('Choose step to adjust, check values or finish setup.'
    )
1159                        order = input('noise / better / fine / hyperfine / check /
    finish')
1160                        if order == 'noise':
1161                            print('Adjusting "noise" position range... (asuming
    min_pos = -max_pos)')
1162                            pos_1 = int(input('max_pos = '))
1163                            print('Adjusting "noise" step length...')
1164                            step_1 = int(input('step length = '))
1165                            print('Adjusting "noise" frequency...')
1166                            freq_1 = int(input('freq = '))
1167                            clear_output()
1168                            print(f'Adjusted parameters for "noise" order: pos:{
    pos_1}, step:{step_1}, freq:{freq_1}')
1169
1170                        elif order == 'better':
1171                            print('Adjusting "better" position range... (asuming
    min_pos = -max_pos)')
1172                            pos_2 = int(input('max_pos = '))
1173                            print('Adjusting "better" step length...')
1174                            step_2 = int(input('step length = '))
1175                            print('Adjusting "better" frequency...')
1176                            freq_2 = int(input('freq = '))
1177                            clear_output()
1178                            print(f'Adjusted parameters for "better" order: pos:{
    pos_2}, step:{step_2}, freq:{freq_2}')
1179
1180                        elif order == 'fine':
1181                            print('Adjusting "fine" position range... (asuming
    min_pos = -max_pos)')
```

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                            pos_3 = int(input('max_pos = '))
                            print('Adjusting "fine" step length...')
                            step_3 = int(input('step length = '))
                            print('Adjusting "fine" frequency...')
                            freq_3 = int(input('freq = '))
                            clear_output()
                            print(f'Adjusted parameters for "fine" order: pos:{pos_3
}, step:{step_3}, freq:{freq_3}')

                        elif order == 'hyperfine':
                            print('Adjusting "hyperfine" position range... (asuming
min_pos = -max_pos)')
                            pos_4 = int(input('max_pos = '))
                            print('Adjusting "hyperfine" step length...')
                            step_4 = int(input('step length = '))
                            print('Adjusting "hyperfine" frequency...')
                            freq_4 = int(input('freq = '))
                            print('Adjusting "hyperfine" iterations...')
                            hyperfine_iterations = int(input('hyperfine_iterations =
 '))
                            clear_output()
                            print(f'Adjusted parameters for "hyperfine" order: pos:{
pos_4}, step:{step_4}, freq:{freq_4}')

                        elif order == 'check':
                            print(f'Noise alignmet: position range: {pos_1}, step
length: {step_1}, frequency: {freq_1} ')
                            print(f'Better alignmet: position range: {pos_2}, step
length: {step_2}, frequency: {freq_2} ')
                            print(f'Fine alignmet: position range: {pos_3}, step
length: {step_3}, frequency: {freq_3} ')
                            print(f'Hyperfine alignmet: position range: {pos_4},
step length: {step_4}, frequency: {freq_4} ')

                        elif order == 'finish':
                            print('Running script')
                            break

                        else:
                            print('Input error')

                    clear_output()
                    break

```

```
                else:
                    print('Input error, allowed inputs: "yes" or "no"')

                # Run automatic stage alignment for given values
                auto_stage_alignment(stage, pos_1, step_1, freq_1, pos_2, step_2, freq_2, pos_3, step_3, freq_3, pos_4, step_4, freq_4, hyperfine_iterations)


        elif ch == "ext":
            # Exit the control script
            print('Exiting COFB control script')
            break

        else:
            # If ch sting is not a know input
            clear_output()
            print(f'"{ch}" is not a valid input, please input one of the programme functions.')

    print('Programme finished.')

    # Sets running to true for continuous script control
    running = True

## Run COFB control script
COFB_control()
```